Zixun Zhou 2026年07月25日

ZIXUN - Investigate BloG

🥚HOW TO READ THIS PADLET 🥚

🥚 INSRTUCTION 🥚

🥚 This blog documents one investigation told in three movements — not three separate projects.
🥚 it opens with BAZI, a dead end I walked far enough into to learn why it was one.
🥚 It passes through the TAVERN DETOUR, a game I built for fun
that quietly handed me the method AI Life would later run on.
🥚 It lands on AI LIFE, the main line, where the strongest work lives.

Every card carries four fields, so you can place it at a glance without reading the body:

- 🥚**FIELD 1 — TYPE:**
  what kind of card this is (question / reasoning / primary / precedent / build / context).
- 🥚**FIELD 2 — DATE:**
  when the thinking happened, so the timeline reads as a real trajectory.
- 🥚**FIELD 3 — ID:**
  where it sits in the chain, e.g. AL-Q4 = AI Life, main question chain, step 4; lets you follow references back.
- 🥚**FIELD 4 — STATUS:**
  standing = where a card sits in the research; solves the 'what is this dg here' problem.

0

🥚FIELD 1--TYPE

colour swatch; the main category = "what this is"

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⬜ CONTEXT

Background setup ('what is Bazi', 'what is Silly Tavern').

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🟧 PRIMARY RESEARCH

First-hand research (field interviews).

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🟨 INSIGHT

Reasoning / reflection / changing my mind — the strongest kind; don't bury it under MAKING.

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🟩 MATERIAL / PROCESS / MAKING

Hands-on building / technical implementation / code files.

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🟦QUESTION

Poses or advances a question.

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🟪 **PRECEDENT**

Literature / precedent / exhibition research.

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🥚 **FIELD 2--ID**

position = "which step; also lets cards cross-reference each other"

ID
BAZI-06

Every card has a short ID, and the ID does two jobs at once. The **prefix** tells you which part of the investigation a card belongs to; the **number** tells you where it sits within that thread. So the ID alone lets you place a card and follow references between cards — when one card builds on another, its ID points you straight back.

The investigation runs in three movements, and the prefixes follow them:
BAZI- — Movement 1, my study of Bazi, the direction I eventually retired. Numbered BAZI-01 to 16 in the order I worked through it.
RP- — Movement 2, the Tavern Detour: a game I built for fun that ended up handing me the method AI Life runs on. Numbered RP-01 to 10.
AL- — Movement 3, AI Life, the main line. Because this is where most of the work is, the numbers break down by *what kind* of card it is:

- **AL-Q** — the main reasoning chain, my spine of questions. It runs Q1 → Q7, and the numbers are the argument: each question grows out of the one before it, so reading Q1 through Q7 in order *is* the thinking.
- **AL-R** — my close reading of one key paper (Nisioti et al.), noted card by card as R1 → R7.
- **AL-C** — a cited source **or a work encountered** (paper, exhibition, publication)
- **AL-I** — a standalone piece of reasoning or reflection (I1, I2…).
- **AL-P** — the field interviews with long-term keepers (P1 → P4; P0 is the interview template).
- **AL-M** — runs continuously in build order across both experiments: M1–M24 nurtured type (Plan B), M25–M36 wild type (Plan A).
- **AL-S** — an overview document.

PR- — the closing section, my reflection on the portfolio itself (PR-01 onward).
RE- — Movement 4, Realise: from experiment to object.

In short: read the prefix to know *what* you're looking at and *which thread* it's on, and read the number to know *which step*. If you only follow one thread, follow **AL-Q1 → Q7** — that's the argument the whole project hangs on.

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🥚**FIELD 3--DATE**
gives the inquiry a time depth

Time
2026.5.5

0

🥚**FIELD 4--STATUS**
standing = "where a card sits in the research; solves the 'what is this doing here' problem"

Status
DEAD END

- **DEAD END:**
  A path I consciously walked and recorded — a plus for Investigate, not a flaw.
- **LOAD-BEARING:**
  A card the argument actually stands on; read these closely. RUNNING Experiment still running / no conclusion yet / bug unresolved — an honest 'in progress'. SUPPORTING: Serves the main line without being it (citations, smoke tests, files, dashboards).
- **SEED:**
  A card that grew into a later direction — the origin of something bigger (the Tavern cards).
- **DONE:**
  A finished build kept as process evidence — neither dead nor running (the cloud-deployment log).

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PROJECT PROPOSAL

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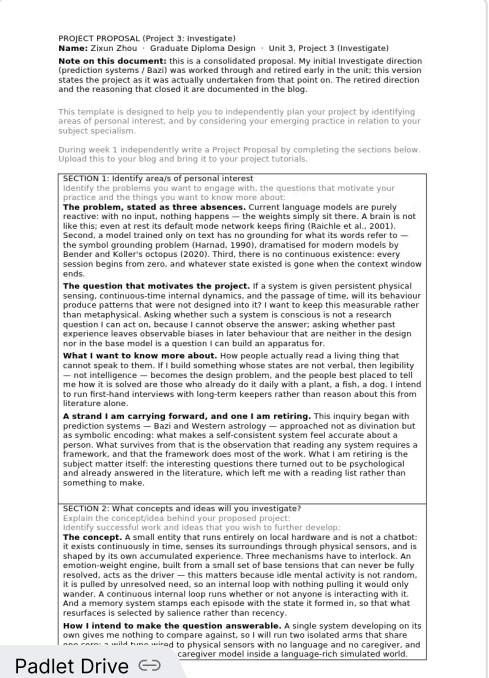
PROJECT PROPOSAL (Project 3: Investigate)
Name: Zixun Zhou · Graduate Diploma Design · Unit 3, Project 3 (Investigate)
Note on this document: this is a consolidated proposal. My initial Investigate direction (prediction systems / Bazi) was worked through and retired early in the unit; this version states the project as it was actually undertaken from that point on. The retired direction and the reasoning that closed it are documented in the blog.
This template is designed to help you to independently plan your project by identifying areas of personal interest, and by considering your emerging practice in relation to your subject specialism.
During week 1 independently write a Project Proposal by completing the sections below. Upload this to your blog and bring it to your project tutorials.
SECTION 1: Identify area/s of personal interest
Identify the problems you want to engage with; the questions that motivate your practice and the things you want to know more about:
The problem, stated as three absences. Current language models are purely reactive: with no input, nothing happens — the weights simply sit there. A brain is not like this; even at rest its default mode network keeps firing (Raichle et al., 2001). Second, a model trained only on text has no grounding for what its words refer to — the symbol grounding problem (Harnad, 1990), dramatised for modern models by Bender and Koller's octopus (2020). Third, there is no continuous existence: every session begins from zero, and whatever state existed is gone when the context window ends.
The question that motivates the project. If a system is given persistent physical sensing, continuous-time internal dynamics, and the passage of time, will its behaviour produce patterns that were not designed into it? I want to keep this measurable rather than metaphysical. Asking whether such a system is conscious is not a research question I can act on, because I cannot observe the answer; asking whether past experience leaves observable biases in later behaviour that are neither in the design nor in the base model is a question I can build an apparatus for.
What I want to know more about. How people actually read a living thing that cannot speak to them. If I build something whose states are not verbal, then legibility — not intelligence — becomes the design problem, and the people best placed to tell me how it is solved are those who already do it daily with a plant, a fish, a dog. I intend to run first-hand interviews with long-term keepers rather than reason about this from literature alone.
A strand I am carrying forward, and one I am retiring. This inquiry began with prediction systems — Bazi and Western astrology — approached not as divination but as symbolic encoding: what makes a self-consistent system feel accurate about a person. What survives from that is the observation that reading any system requires a framework, and that the framework does most of the work. What I am retiring is the subject matter itself: the interesting questions there turned out to be psychological and already answered in the literature, which left me with a reading list rather than something to make.
SECTION 2: What concepts and ideas will you investigate?
Explain the concept/idea behind your proposed project:
Identify successful work and ideas that you wish to further develop:
The concept. A small entity that runs entirely on local hardware and is not a chatbot: it exists continuously in time, senses its surroundings through physical sensors, and is shaped by its own accumulated experience. Three mechanisms have to interlock. An emotion-weight engine, built from a small set of base tensions that can never be fully resolved, acts as the driver — this matters because idle mental activity is not random, it is pulled by unresolved need, so an internal loop with nothing pulling it would only wander. A continuous internal loop runs whether or not anyone is interacting with it. And a memory system stamps each episode with the state it formed in, so that what resurfaces is selected by salience rather than recency.
How I intend to make the question answerable. A single system developing on its own gives me nothing to compare against, so I will run two isolated arms that share one core: a wild type wired to physical sensors with no language and no caregiver, and caregiver model inside a language-rich simulated world.
Padlet Drive

Investigate_Project_Proposal_ZixunZhou

INVESGATE DIRECTION 1: BAZI

Movement 1 — Bazi (May 2026, retired)
This section is a direction I worked through and then left. It runs BAZI-01 to BAZI-16 in the order I actually thought about it, so the drift is visible: it opens as a question about whether a prediction system is true, and it ends somewhere else entirely — at why people need one whether or not it is.

Every card here carries the status DEAD END. In this Padlet that is a position, not a verdict on the work. It means a path I walked far enough to know why I was leaving it, and the reason is stated on the card that closes this section rather than assumed.

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It all started with a stupid question: Neptune has no ocean — so why does it represent ocean energy?

https://www.amazon.com/History-Western-Astrology-Ancient-Classical/dp/1441127372

This question hit me one day and felt genuinely absurd. Astrology says Neptune represents the ocean, fluidity, and unfathomable depth — but Neptune's actual composition is methane, ammonia, and icy mixtures, with surface temperatures around -214°C. Nothing remotely oceanic. So where does this "energy" come from? The answer turned out to be simple: when astronomers discovered the planet in 1846, they named it after the Roman sea god Neptune by convention. The astrological system then took that name's mythological imagery — sea god, ocean, fluidity — and packaged the whole thing as the planet's "energy attributes." That's it. No deeper reason. This made me realise something: the "planetary energies" in astrology are fundamentally mythological associations, not properties derived from astrophysics. If that planet had been named "Forest Goddess," astrologers today would say it represents nature, vitality, and green energy — and it would sound equally convincing. The system's validity doesn't depend on what the starting point is — it depends on the self-consistent elaboration that follows once the starting point is fixed. That felt like an important finding.

DATE

2026.5.5

ID

BAZI-01

Status

DEAD END

0

What is astrology's logic — it was never a physical system to begin with

https://www.amazon.com/History-Western-Astrology-Ancient-Classical/dp/1441127372

After the Neptune realisation, I started asking: what is astrology's actual operating logic? What's its basis for claiming planetary positions affect personality and destiny? The answer: it has no such basis, because it was never describing physical reality in the first place — it operates a symbolic system. Western astrology's core framework was established in the Greco-Babylonian period, when planets were just moving points of light. Nobody knew Jupiter was a gas giant or Mars a red desert. What practitioners had was: visible colour (Mars appears red → blood and war), mythology (this star is named the war god Mars → its energy is war and force), and accumulated observational impressions. Once the framework crystallised, it became self-enclosed. When Uranus, Neptune, and Pluto were discovered, astrologers simply inserted the mythological imagery of their new names — the framework itself never updated. So astrology's logic chain runs: planet → corresponding mythological figure → symbolic meaning → energy attributes. Not: planet → physical properties → actual effects on human bodies. These are parallel language systems with fundamentally different starting points. Astrology never intended to derive energy attributes from astrophysics — it uses mythological archetypes as projection tools. Once you understand this, "why does astrology say X" always has the same answer: because that's what the myth says.

DATE

2026.5.5

ID

BAZI-02

Status

DEAD END

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So why does Bazi hold up better than astrology — derived from astrology's gaps

After thinking through astrology's problems clearly, I applied the same standards to Bazi and found its underlying logic substantially more grounded — not because Bazi is necessarily accurate, but because its physical core is more “real”.
Astrology's central leap: the positions of distant planets at the moment
of birth affect your personality. No physical mechanism exists for this.
Bazi's core chain:
birth time → corresponding season and month → actual foetal developmental environment (temperature, light exposure, maternal nutrition) → neural development and temperament tendencies → behavioural patterns → life trajectory.
The first several steps in this chain have genuine biological basis. Modern research has found statistically significant correlations between birth season and personality traits. So Bazi captured a real signal: birth time genuinely carries information about developmental environment, and developmental environment genuinely affects people.
Of course Bazi then built an elaborate symbolic superstructure on this core — Heavenly Stems, Earthly Branches, Five Elements, Luck Pillars — and whether that superstructure can support the level of predictive precision claimed is a separate question. But at least its starting point is more grounded than "sea god represents the ocean." This comparison is what made me decide to investigate Bazi rather than Western astrology.

DATE
2026.5.5

ID
BAZI-03

Status
DEAD END

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What is Bazi? It's not fortune-telling — it's an encoding system.
When I first encountered Bazi, I dismissed it as superstition. But breaking it down revealed a surprisingly rigorous internal logic. The "eight characters" (八字) encode a person's birth time into Heavenly Stems and Earthly Branches across four pillars — year, month, day, and hour. These eight characters map the relationships and relative strengths of the Five Elements (Wood, Fire, Earth, Metal, Water) at the moment of birth. Crucially, Bazi doesn't jump directly from birth time to destiny — there's an entire calculation system in between: Five Element interactions, Ten Gods relationships, Luck Pillars and Annual Pillars. Each step follows defined rules. I now think of it as a multivariate modelling system built in ancient symbolic language: time coordinates go in, Five Element dynamics are processed, and personality tendencies and life trajectory come out. Whether this encoding accurately reflects reality is a separate question — but the logical architecture itself is structurally identical to modern data modelling.

DATE
2026.5.6

ID
BAZI-04

Status
DEAD END

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Bazi is a multivariate system — not as crude as "born in winter = introverted"
I initially made the mistake of thinking Bazi was just "your birth month determines your personality." Going deeper revealed something far more complex — and that complexity is precisely where its value lies. A Bazi chart has four pillars, each containing a Heavenly Stem and Earthly Branch, with hidden stems (藏干) concealed within each branch. That alone creates a system with eight to over a dozen variables. These variables interact through Five Element generating and overcoming cycles — Wood feeds Fire, Fire feeds Earth, and so on, alongside the overcoming relationships. On top of that, Luck Pillars (大运, changing every ten years) and Annual Pillars (流年) layer additional variables into the calculation. So when people say "Bazi doesn't work," the problem might not be the system itself — it might be that most practitioners lack the skill to operate a genuinely complex multivariate model. Like a sophisticated statistical framework with sound methodology but inconsistent operators.

DATE
2026.5.6

ID
BAZI-05

Status
DEAD END

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The Barnum Effect: why vague descriptions feel like they're talking about you specifically

This is the core psychological mechanism explaining why fortune-telling, astrology, and tarot feel accurate. In 1949, psychologist Bertram Forer gave each of his students a personality test, then handed them what appeared to be a personalised analysis. Students rated these analyses as highly accurate — but everyone received the same text, which Forer had copied from a newspaper astrology column. This is the Barnum Effect: as long as a description is sufficiently vague and positive, most people will feel it describes them personally. The human brain actively searches for matches between ambiguous information and personal experience, remembers the hits, and forgets the misses (confirmation bias). This doesn't mean Bazi is fraudulent — it means that even if Bazi descriptions were entirely random, many people would still find them accurate. It's an inherent cognitive bias, not proof of Bazi's validity or its refutation. It simply explains why "does it feel accurate to users" is a terrible way to evaluate whether a system actually works.

DATE
2026.5.10

ID
BAZI-06

Status
DEAD END

0

Why people need prediction systems — anxiety and uncertainty are the real drivers

A research finding struck me as particularly sharp: once you control for emotional need variables — anxiety levels, desire for control, and tolerance for uncertainty — educational background has almost zero predictive power over whether someone believes in astrology or fortune-telling. Highly educated people and credulous ones aren't that different when it comes to this. This suggests that belief in prediction systems is fundamentally not an intellectual issue but a psychological need. When people face questions like "what will happen to me" and "do I have a destiny," genuine anxiety is produced. Prediction systems — whether Bazi, astrology, or tarot — provide a structured narrative framework that transforms "the future" from chaos into something comprehensible. This function works regardless of predictive accuracy, because what it satisfies is the psychological need to feel seen, understood, and in control. Notably, usage rates surged during the pandemic, especially

among young people. That's not coincidental — when external uncertainty spikes, people need some system to re-establish a sense of order.

DATE

2026.5.10

ID

BAZI-07

Status

DEAD END

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The ritual of consulting a fortune-teller has value independent of accuracy

This challenged my assumptions: sitting down with a fortune-teller creates a distinctive cognitive state — different from scrolling your phone or casual conversation. The ritual signals "this moment matters," forces you to slow down, and focuses attention on your own situation and choices. From this angle, fortune-telling functions more as a structured self-reflection tool than purely a prediction mechanism. The fortune-teller's questions — "has work been going well recently?" "any changes in relationships?" — systematically guide you through reviewing your own life, something most people never do deliberately. So regardless of what's predicted, the process itself already performs something resembling psychological counselling. This connects to Jung's analysis — he believed astrology operates not through physical causation but as a "synchronicity" tool, helping people draw on collective unconscious archetypes to understand their situation. Whether or not you buy Jung, this functional reading aligns with what I've observed.

DATE

2026.5.10

ID

BAZI-08

Status

DEAD END

0

EYES 1: The illusion of control: why "knowing your fate" paradoxically makes people feel more in control

📎 Sosis, R.H., Strickland, B.R., & Haley, W.E. (1980). "Perceived locus of control and beliefs about astrology." *Journal of Social Psychology*, 110(1), 65–71. → https://www.tandfonline.com/doi/abs/10.1080/00224545.1980.9924223
📎 Wunder, E. (1998). "Belief in astrology as a strategy for self-verification and coping with negative life-events." *European Psychologist*, 3(3), 202–208. → https://econtent.hogrefe.com/doi/abs/10.1027/1016-9040.3.3.202

This is the most counter-intuitive finding in the psychological layer. Logically, "your destiny is fixed" should induce feelings of helplessness — but research shows the opposite: people who believe in fate-based systems often score higher on certain measures of perceived control. The reason: the sense of control doesn't come from "I can change outcomes" but from "I can understand what's happening." When someone is in crisis or chaos, the worst feeling isn't the outcome itself but the bewilderment of "why is this happening" and "what comes next." Bazi or astrology provides an interpretive framework — even if symbolic and external — and as long as it transforms chaos into "readable," people experience psychological relief. One experimental study had participants learn astrology, finding that astrology students (compared to psychology or German language control groups) showed significantly increased certainty about self-descriptive attributes and heightened perceptions of having control. Not because astrology changed reality — but because it gave people a language to understand themselves.

DATE
2026.5.10

ID
BAZI-09

Status
DEAD END

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EYES 2:
Self-concept verification: people use Bazi and astrology to confirm "who I am"

Psychology has a concept called Self-Verification Theory: people have a fundamental need for external feedback to be consistent with their self-concept — whether that concept is positive or negative. In other words, we don't just want to be praised; we want to be accurately seen. This mechanism is particularly visible in how people use Bazi and astrology. When someone sees a reading, the first response is often not "does this match my situation?" but a feeling of recognition: "this is describing me." Research shows astrology is especially appealing in ambiguous or uncertain situations — precisely when people most need an external system to stabilise their self-concept. This also explains why tools like MBTI, star signs, and Bazi are so popular among young people today: they offer not just "what type of person are you" but a sense of identity that has been certified by an authoritative system. "I'm an INFJ," "I'm a Scorpio," "I'm a Jiǎ Wood Day Master" — these labels also serve social functions: shortcuts for self-introduction, tools for establishing common ground with strangers.

📎 Self-Verification Theory overview: https://www.simplypsychology.org/self-verification-theory.html
📎 Wunder (1998) — 占星作为自我验证策略的实验研究：https://econtent.hogrefe.com/doi/abs/10.1027/1016-9040.3.3.202
📎 *From Astronomy to Astrology: Testing the Illusion of Zodiac-Based Personality Prediction with Machine Learning*, arXiv 2026: → https://arxiv.org/html/2603.29033v1

DATE
2026.5.10

ID
BAZI-10

Status
DEAD END

0

Confirmation bias: why you only remember when it was right

📎 Nickerson, R.S. (1998). "Confirmation bias: A ubiquitous phenomenon in many guises." *Review of General Psychology*, 2(2), 175–220. → https://doi.org/10.1037/1089-2680.2.2.175
📎 *The Barnum Effect* — The Decision Lab综述，含确认偏误与记忆选择性的讨论： → https://thedecisionlab.com/biases/barnum-effect

This is the memory mechanism version of the Barnum Effect, worth addressing separately because it explains something very specific: why fortune-telling feels subjectively "far more accurate than it actually is." Human memory isn't a video recording — it's reconstructive and selective. When someone believes in a prediction system, the brain automatically executes two operations: first, actively searching for evidence that matches the prediction (the hits); second, reducing attentional weight on non-matching evidence and forgetting it more rapidly. This is confirmation bias operating at the memory level. So a fortune-teller says ten things, three vaguely match, seven completely miss — you remember the three, you tell your friend "she was so accurate." You're not lying; this is how the brain normally operates. Research shows information consistent with existing beliefs is selected and remembered at nearly twice the rate of inconsistent information. This mechanism makes "word-of-mouth validation" of Bazi or astrology almost entirely unreliable as a system evaluation method — even if the system is completely invalid, transmitted accuracy rates will be inflated.

DATE
2026.5.10

ID
BAZI-11

Status
DEAD END

0

External locus of control and fate belief: who is most likely to believe in fortune-telling

Not everyone is equally receptive to fortune-telling. Psychological research has identified a stable correlation: external locus of control personality orientation is significantly positively correlated with fortune-telling belief. "Locus of control" is a personality psychology concept proposed by Julian Rotter in 1954: internally oriented people believe their own actions and abilities determine life outcomes; externally oriented people believe outcomes are primarily controlled by external forces (fate, powerful others, chance). Research shows that the stronger someone's external orientation, the more they believe in and actively use astrology or fortune-telling. But there's something subtle here — this doesn't mean external locus of control is a defect. It more likely reflects the fact that when people genuinely feel "many things are outside my control," turning to external interpretive systems is a natural response. The surge in astrology use during the pandemic, the consistently higher usage rates among women (who historically have had less structural power in society) — all of this aligns with this mechanism. This isn't superstition; it's an adaptive response to specific circumstances.

📎 Sosis, R.H., Strickland, B.R., & Haley, W.E. (1980). "Perceived locus of control and beliefs about astrology." *Journal of Social Psychology*, 110(1), 65–71. → https://www.tandfonline.com/doi/abs/10.1080/00224545.1980.9924223
📎 Rotter, J.B. (1966). "Generalized expectancies for internal versus external control of reinforcement." *Psychological Monographs*, 80(1). → Wikipedia Locus of Control综述：https://en.wikipedia.org/wiki/Locus_of_control

DATE
2026.5.10

ID
BAZI-12

Status
DEAD END

0

Jung's reframe: astrology is not physical causation — it's a psychological projection tool

Jung was one of the few academic psychologists of the 20th century who took astrology seriously. But his defence of it was completely different from ordinary believers' — he didn't think planets physically affect people, but that astrology is a psychological projection system, a tool for accessing collective unconscious archetypes. His core concept is "synchronicity": certain events have meaningful coincidental connections without causal relationships. A planet's position doesn't cause your personality — rather, it "resonates" with your psychological state at a given moment. This resonance is meaningful and worth interpreting, but it's not a physical mechanism. More importantly, his view on archetypes: the zodiac signs and major planets have accumulated over thousands of years into archetypal symbols in the human collective unconscious — Mars doesn't represent the planet itself but the shared human psychological archetype of "aggression/drive/competition." When you "read" yourself through this system, you're borrowing an ancient symbolic language to access inner content that would otherwise be difficult to articulate directly. This framework also makes Bazi coherent: the Heavenly Stems, Earthly Branches, and Five Elements aren't

physical prediction tools, but a sufficiently sophisticated symbolic language that helps you approach self-understanding from different angles.

📎 Jung, C.G. (1960). *Synchronicity: An Acausal Connecting Principle*. Princeton University Press. ISBN: 9780691150505 → Goodreads书目页：https://www.goodreads.com/book/show/836937
📎 Jung on Astrology — Archetypes and the Psyche综述： → https://natalecho.com/learn/astrology/carl-jung-and-astrology-archetypes-synchronicity
📎 Mindberg — Jung and Astrology: Psychological Insights： → https://mindberg.org/astrology-psyche-jung/

DATE
2026.5.10

ID
BAZI-13

Status
DEAD END

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The meaning vacuum: as religion declines, people still need answers to "why"

This is a sociologically interesting finding. One of the core functions traditional religion provides is a "why" explanation for personal experience: why did this happen to me? Does my suffering have meaning? Do I have a place in the universe? But in Western countries, and among younger generations in China, religious belief is declining rapidly. US data from 2019 shows self-identified "religious nones" rose from 17% in 2009 to 26%. A 2023 UK survey found the proportion of the public identifying as theistic fell from 75% in 1981 to 49%. The problem: religious decline doesn't mean the need for meaning disappears. Research finds that "the need for meaning" independently predicts both religiosity and belief in paranormal phenomena (including astrology and fortune-telling) — both predictions hold after controlling for other variables. In other words, Bazi and astrology are substantially filling the "meaning explanation" functional gap vacated by religion. This doesn't mean everyone using Bazi is doing religious substitution — it means that at the social structural level, when traditional meaning supply decreases, people seek it from other sources. Co-Star astrology app's growth from 7.5 million users in 2020 to 30 million in 2023 is not coincidental.

📎 FioRito, T.A., Abeyta, A.A., & Routledge, C. (2021). "Religion, paranormal beliefs, and meaning in life." *Religion, Brain & Behavior*, 11(2), 139–146. → DOI: https://doi.org/10.1080/2153599X.2020.1824938 → Taylor & Francis全文: https://www.tandfonline.com/doi/full/10.1080/2153599X.2020.1824938
📎 Oxford Brookes University (2025). "Why is Gen Z looking to the stars for meaning?" → https://www.brookes.ac.uk/about-brookes/news/news-from-2025/05/expert-comment-why-is-gen-z-looking-to-the-stars-f

DATE
2026.5.10

ID
BAZI-14

Status
DEAD END

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Astrology and Bazi as social currency — you don't have to believe it; it's a language for connection

This dimension is different from all the psychological need layers above — it's purely social function. Observe reality: large numbers of young people use star signs and Bazi without necessarily believing it can predict destiny. They use it more like a social shorthand language — "I'm a Scorpio" conveys far more information than a simple self-introduction, implying a personality profile and giving the other person a quick framework for understanding you. "What's your star sign?" is a low-cost icebreaker; once both parties use this language, a form of shared cultural belonging is established. Research shows TikTok has over 4.5 million videos tagged "astrology," and astrology content has permeated fashion, beauty, and brand marketing on Instagram. Psychologist Anirudh Rao notes: "Memes allow astrology to function as both identity and humour. A person might not deeply believe in star signs but still share a 'Leo confidence' meme because it feels relatable or funny. That's how astrology enters everyday conversation." Bazi's spread among young Chinese people follows similar mechanisms — "checking compatibility" (查合婚) serves the social function of creating a shared ritual for two people to examine their relationship together, not necessarily making destiny predictions. The ritual itself creates intimacy.

📎 Newsmeter (2025). "Astrology finds new life on Instagram as millennials and Gen Z use it for identity, community." → https://newsmeter.in/whats-new/astrology-finds-new-life-on-instagram-as-millennials-and-gen-z-use-it-for-identity-community-754657
📎 Oxford Brookes (2025) — "Community and belonging: Much like religious congregations, astrology fosters strong online and offline communities." → https://www.brookes.ac.uk/about-brookes/news/news-from-2025/05/expert-comment-why-is-gen-z-looking-to-the-stars-f

DATE
2026.5.10

ID
BAZI-15

Status
DEAD END

0

Fortune-telling as a "permission mechanism" — finding external authorisation for decisions you've already made

This is the most subtle and practically interesting function — I think many people using Bazi or fortune-telling are actually doing this without necessarily being aware of it. The scenario: a person is already internally leaning toward a decision (changing jobs, ending a relationship, moving), but because the choice carries risk or would face opposition, they need some external authorisation to give themselves permission. So they consult a fortune-teller: if the result supports their inclination, they feel "permitted" and can act; if not, they might find a second fortune-teller until they get the answer they want. Research confirms fortune-telling's real impact on decision behaviour. A study published in PLOS ONE found that even among people who self-identify as non-superstitious, men's financial risk-taking behaviour significantly increased after receiving a positive fortune-telling prediction — researchers explained this as the result providing psychological authorisation to "have a reason to take the risk," reducing decision anxiety. A separate qualitative study of Chinese users of AI fortune-telling tools found users divided into two types: those seeking entertainment and psychological comfort had higher acceptance; those making major life decisions treated readings as "supplementary input rather than authoritative guidance." This distinction is itself interesting: people use the "permission mechanism" differently depending on context.

📎 Tan, X., van Prooijen, J-W., & van Lange, P.A.M. (2022). "Positive fortune telling enhances men's financial risk taking." *PLOS ONE*, 17(9): e0273233. → PMC全文: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9451074/
→ DOI: https://doi.org/10.1371/journal.pone.0273233
📎 "Re-Tell the Fortune so I Can Believe It": How Chinese User Communities Engage with and Interpret GenAI-based Fortune-Telling, arXiv 2025. → https://arxiv.org/pdf/2603.27784

DATE
2026.5.10

ID
BAZI-16

Status
DEAD END

0

Why I left this direction

Sixteen cards in, I noticed what had happened to the question rather than to the answer.

I started by asking whether these systems hold up. BAZI-01 to BAZI-05 are that question, and they reach an answer I still stand behind: astrology's starting point is mythological convention (a planet named after a sea god is therefore said to carry the ocean), Bazi's starting point is at least anchored in something physical (birth season really does correlate with developmental environment), and both then elaborate a self-consistent symbolic system on top of that starting point. The validity of such a system does not depend on where it starts — it depends on the consistency of what follows.

But from BAZI-06 onward the question had quietly changed underneath me. Barnum effect, confirmation bias, locus of control, the meaning vacuum, fortune-telling as permission mechanism — every one of those cards is about why people need a prediction system. I do not disown them; they are good findings.

The problem is what kind of findings they are. Every one is already answered in existing psychology literature, and not one of them gave me anything to build. I had turned a making project into a reading list.

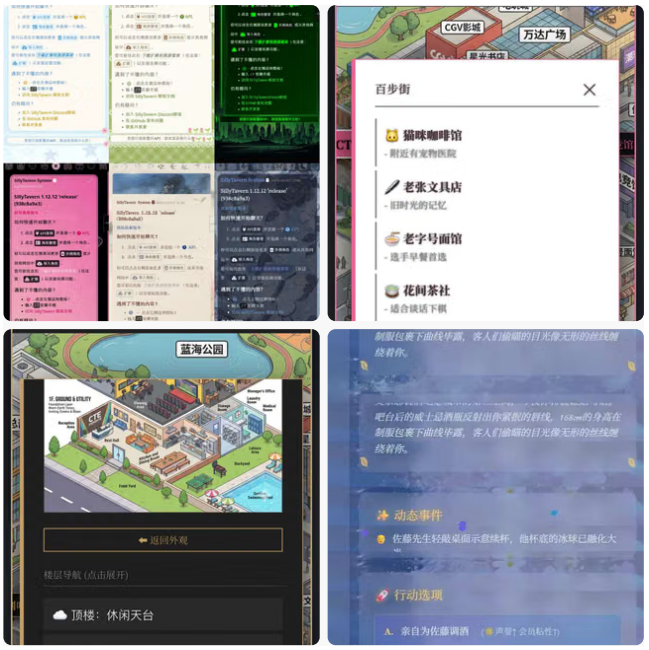
That is the actual reason this is a dead end, and it is more useful to me than the content of any single card in this section, because it is a failure mode I can now recognise early: a line of inquiry that only ever produces more reading is not a line of inquiry for this unit.

What I moved to next was not a better version of this question. It was RP-01 — a game I had already been playing for fun, which turned out to have a method inside it.

0

🐔INVESGATE DIRECTION 2: AI ROLE PLAY🐔

WHAT IS SILLY TAVERN?



Silly Tavern is an open-source role-playing dialogue tool that allows me to engage in immersive, game-like interactions with AI characters. More than just a chat interface, it offers robust customization features, such as adjusting character personalities, building multi-character interaction scenarios, and connecting to different large language models. This tool has made me rethink the potential of "conversational creation" — instead of being a one-way creator, I now collaborate with AI to dynamically advance stories and explore characters through interactive exchanges. This collaborative creative technique has completely transformed my previous linear and isolated approach to making.

DATE

2026.3.10

ID

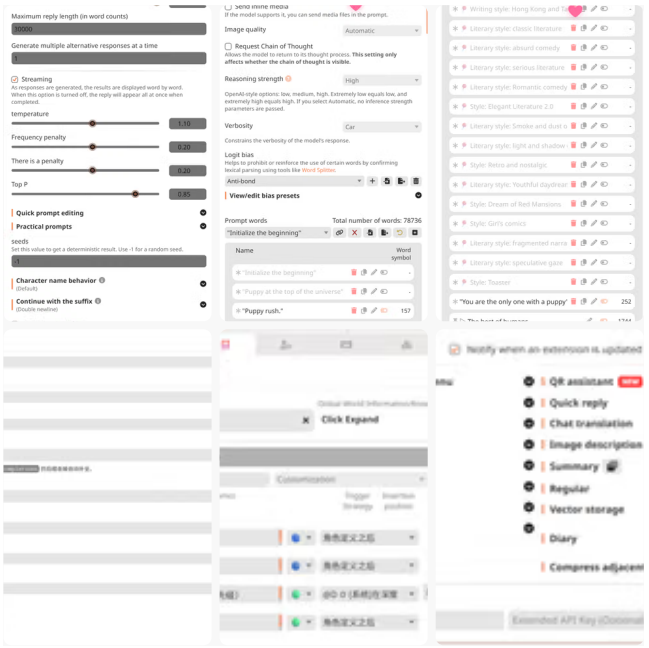
RP-01

Status

SEED

0

From API Configuration to Front-End Possibilities: A Tavern AI Research Log



I first came across the concept of Tavern half a year ago. I have a great passion for RPGs, D&D and wargames, so initially, I was drawn to its highly open-ended gameplay format. When I configured the API for the first time, what piqued my curiosity the most was the respective impacts of the World Book, presets, regular expressions and temperature on the AI. I could sense that their effects overlap to a large extent, much like how exposure, shutter speed and other parameters in a camera complement one another.

Thus, my first experiment was to explore how different presets influence the same AI model. I then moved on to researching the World Book, and a key insight emerged: all text-based elements – the World Book, prompts, presets, user inputs, character cards and so on – are essentially the same in nature. The only differences lie in **when** specific information is sent to the AI, the **depth** of that information delivery, and its **position and sequence** in the process. This aspect touches on the core internal

mechanics of Tavern. The division of these elements into distinct modules on the front end is merely a categorical arrangement designed to help us input the right information in the right places.
After gaining a quick and general understanding of these fundamentals, I began investigating the visual and formal possibilities that can be unlocked for AI models through prompts – in other words, the front-end content of the system.

(UPDATED)
Why this card is the method, not just a log:
I wrote the paragraph above about a game, and at the time I thought it was a note about how Tavern's front end happens to be organised. It is the operating principle the whole of AI Life later runs on, and I only recognised that retrospectively — three separate times, each time as a bug.
AL-M9 — the worldbook is not "setting". It is a system prompt. Position and depth are the only things that make it act on anything at all.
AL-M15 — I discovered WORLDBOOK had been imported but never actually placed into the prompt. Every character detail I had written upstream had exactly zero effect on what got generated, because nothing had been *delivered*. That is the claim above, failing in the most literal way available.
AL-M18 — the bilingual rule was present, but buried inside a much longer message, and the model stopped obeying it. Identical information, worse position, different behaviour.
So "all text elements are the same in nature; only *when*, *how deep*, and *in what position* differ" is not an observation about a piece of hobby software. It is the thing I have been debugging against ever since. I had it written down four months before I understood what I had written.

DATE
2026.3.10

ID
RP-02

Status
SEED

0

Exploration Process of Cloud Tavern Construction

Exploration Direction

Build a lightweight cloud-native Silly Tavern interactive system based on the Huawei Cloud Flexus L server and 1Panel visual panel. Explore the technical implementation and iterative problem-solving paths for cloud service deployment, visual panel operation and cross-terminal access adaptation, with a focus on the integration of technical practice and innovative application development.

Exploration Phase

Cloud service basic configuration → Core application deployment → Network link construction → Troubleshooting and system optimization

DATE

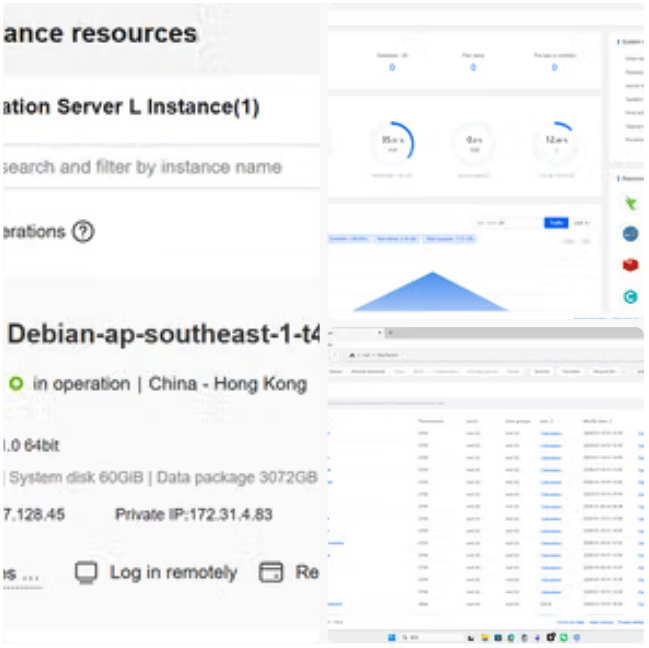
2026.3.10

ID

RP-03

Status

DONE



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Network and Access Issues

1. After the completion of the Cloud Tavern construction, external network access was unavailable. The system could only be accessed normally locally on the server or within the LAN, and the external network access link of the core application Silly Tavern was completely blocked, making it impossible to use remotely across terminals. 2. Port mapping failed after configuration. The corresponding ports have been opened in the Huawei Cloud background, but the port mapping rules configured in the 1Panel panel did not take effect. Some ports showed an open status but could not actually achieve network communication. 3. There were intermittent freezes and connection interruptions during external network access. Although the application could be accessed successfully, the stability was poor, and connection failures or excessively high response delays occurred frequently.

DATE

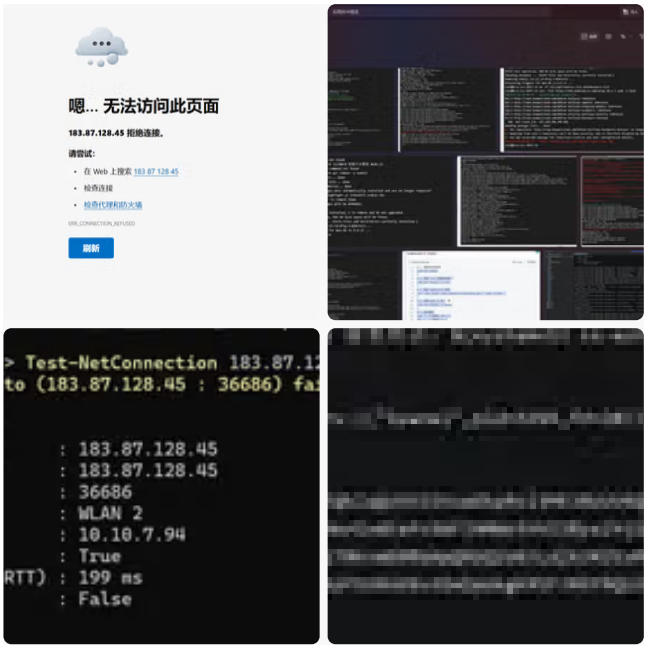
2026.3.10

ID

RP-04

Status

DONE



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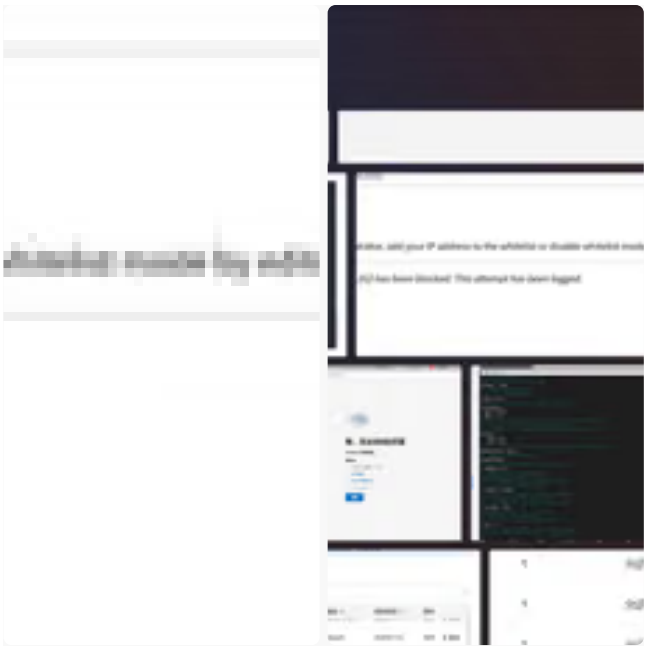
1Panel Panel Operation and Adaptation Issues

1. When deploying Silly Tavern in the 1Panel panel, the application dependency installation failed. Some core plugins and runtime libraries could not be downloaded or configured normally, resulting in the failure to start the application after installation.
2. Insufficient permission errors were frequently encountered during panel operations. When modifying files and installation directories related to Silly Tavern or executing startup commands, normal operations could not be performed due to improper permission configuration, and configuration files could not be saved or take effect.
3. Unfamiliar with the operation of advanced functions of the 1Panel panel, the entrances of key functions such as reverse proxy configuration, environment variable setting, and application monitoring were difficult to locate, making it impossible to complete precise personalized configuration adjustments, which affected the efficiency of application deployment and optimization.



DATE

2026.3.10

ID

RP-05

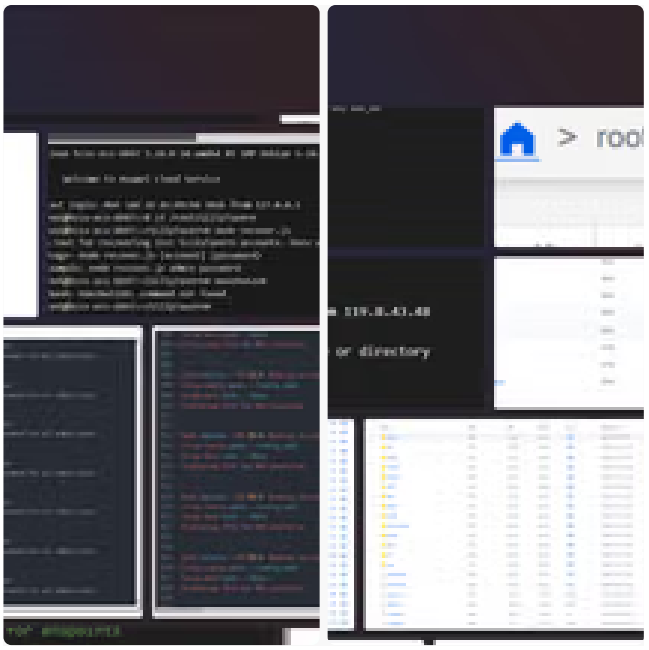
Status

DONE

0

Silly Tavern Core Application Issues

1. The application startup was abnormal. The 1Panel panel showed "Starting" or "Startup Failed" status, but no clear error prompt was given, or the error information was vague, making it impossible to quickly determine the core reason for the startup failure and increasing the difficulty of troubleshooting.
2. Some functional modules failed to load. The basic chat interaction function could be used normally, but extended functions (such as character library loading, conversation memory retention, and third-party plugin calls) could not run normally, and the modules got stuck or directly failed when loading.
3. There was a version adaptation conflict between the application and the server system environment. The version of the runtime environment (such as Python, Node.js)自带 by the server did not match the adaptive version required by Silly Tavern, resulting in compatibility errors or even crashes during the application operation.



DATE

2026.3.10

ID

RP-06

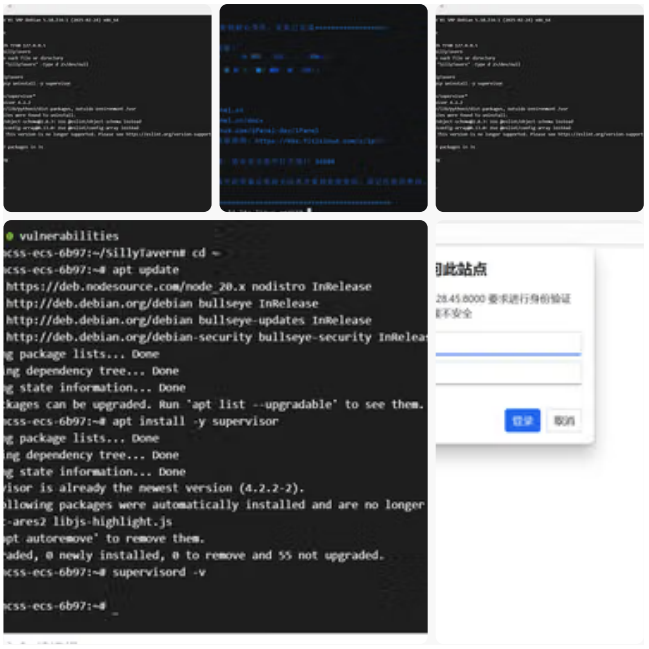
Status

DONE

0

Core Exploration Actions & Progress

1. Completed the initial configuration of the Huawei Cloud Flexus L (monthly subscription) server and successfully installed the 1Panel visual management panel, establishing a basic operation framework for cloud services. This practice realized an efficiency upgrade from pure command-line operation to visual engineering management, exploring a practical path for the lightweight management of cloud services.
2. Deployed the core Silly Tavern application based on the 1Panel panel, sorted out and built the basic dependency environment required for application operation, and initially constructed the core interactive function framework of the Cloud Tavern. This step achieved a key leap from basic cloud service construction to actual application implementation.



DATE
2026.3.10

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RP-07

Status
DONE

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3. Carried out the practice of building an external network access link for the Cloud Tavern, attempted to configure network parameters and open application ports on both the Huawei Cloud background and the 1Panel panel, and promoted the functional expansion from local/LAN operation to cross-terminal external network access, which exposed the core technical challenges in cloud service deployment.

4. Focused on the full-link problems in deployment and operation, and established a systematic troubleshooting thinking of "log investigation - hierarchical verification - comparison and rollback". Specifically, problems were located from multiple dimensions including 1Panel panel logs, server system logs and application local logs; the scope of problems was narrowed step by step in the order of server bottom layer → network layer → panel layer → application layer; and core technical problems such as network configuration, permission adaptation, version compatibility and resource occupation were accurately identified by comparing with official standard processes, verifying local/external network tests, and rolling back key steps.

5. Completed targeted iterative optimization for the identified problems: Restructured the port/firewall rules of the Huawei Cloud security group and the 1Panel panel to solve the problems of external network access failure and port mapping invalidation; Switched software sources, manually supplemented dependencies and adjusted file permissions to resolve dependency installation and permission errors of the 1Panel panel; Built a dedicated multi-version operating environment for the application to address the version compatibility issue between Silly Tavern and the server system; Optimized server resource configuration and application operation parameters to solve abnormal resource occupation and access stability problems, realizing a key iteration of the Cloud Tavern system from startup abnormality to basic stable operation.

DATE
2026.3.10

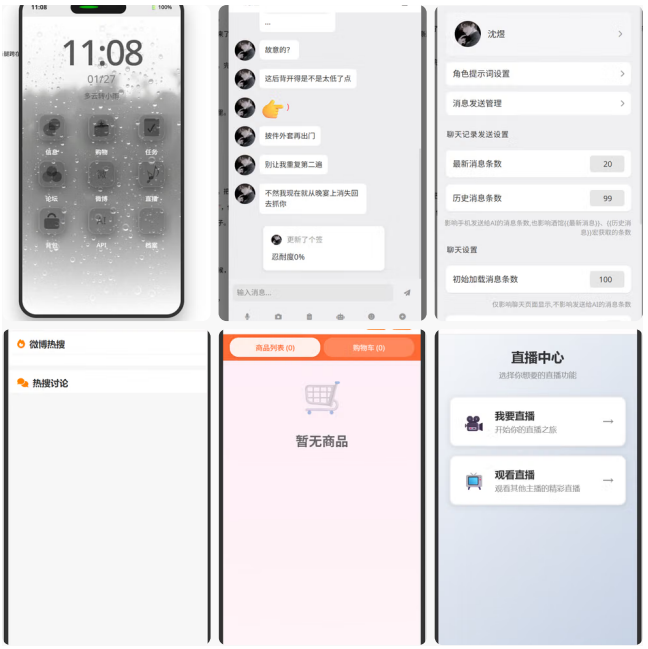
ID
RP-07

Status
DONE

0

Floating Mobile Phone Component

As one of the core interactive entrances, the floating mobile phone component is designed to always float on the edge of the page (drag-adjustable position), with an appearance simulating a real mobile phone interface, deeply associated with the main content of the Cloud Tavern. After clicking to expand, it can realize the integrated use of four core functional modules, fully adapting to scenario-based needs.



DATE
2026.3.10

ID
RP-08

Status
DONE

0

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1. Full WeChat functions: Support simulating friend chats (can be associated with roles in the tavern), group chat interactions, and Moments publishing (content can be synchronized with tavern dynamics). Message reminders are linked with the main content scenarios, for example, the floating mobile phone automatically pops up a prompt when receiving a role reply.

2. Music playback function: Built-in music library, supporting single playback, list loop, and volume adjustment. Background music can be switched according to the tavern scene (such as quiet chat, lively interaction), and the playback status is displayed in real time on the mobile phone floating icon.

3. Notepad function: Support quickly recording important information in the tavern (such as key points of role dialogues, scene plot nodes, to-do items). Recorded content can be exported and synchronized to the tavern personal center, and supports quick access via shortcut keys without affecting the main content browsing operation.

4. Taobao shopping software embedding: Simulate shopping scenarios, browse virtual products related to the tavern theme (such as tavern decorations, character props), support adding to cart and simulating payment processes. Shopping records can be retained in the personal center, forming a linkage with the main content plot (such as unlocking new tavern scenes after purchasing props).

5. Simulated live broadcast/watching live broadcast: Support switching between two modes - the anchor mode can simulate starting a live broadcast in the tavern (displaying scene pictures, real-time interaction), and the audience mode can watch virtual live broadcasts in the tavern (such as role special interaction, scene event live broadcast). The live broadcast comment area is interconnected with the main content chat area to enhance the scene atmosphere.

DATE
2026.3.10

ID
RP-08

Status
DONE

0

FIELD RESEARCH

How the field instrument was built, and what is wrong with it

What I wanted to find out. Not "would people like this device." The question underneath the whole project is whether a system can develop states that are legible to a person without language mediating them (AL-Q7, and the problem I name in RE-01). Before building an expression layer I wanted to know how

people who already do this — who read a creature that cannot speak to them — actually do it. Specifically: what is the *first* signal they notice, is that ability innate or learned, and what does it cost them to learn it.

Why these six questions, in this order.

Q1 is a warm-up and does one quiet job: it establishes whether the respondent relates to one individual or to a group, which turned out to matter more than anything else I designed for (see AL-I[n]).

Q2 is the core. I ask for *the last time*, one concrete story, and I push on the first detail — seen, heard, smelled, touched, or "just felt". The reason for that granularity is that my sensor array is a set of channels, and I wanted to know which channel a human actually reads first.

Q3 asks whether reading is innate or learned, because if it is learned then my device has a second problem it cannot solve alone: the person also has to develop.

Q4 asks for a moment of non-verbal expression that was understood. This is the closest thing to a direct test of the project's premise using an existing, non-designed creature.

Q5 is the speculative probe — I describe my own entity and ask for the first word that comes to mind.

Q6 is an open exit, on the standard assumption that the best material arrives after the instrument stops asking.

Disclosure: I wrote this instrument with an AI assistant (Claude). I set the research question, decided what each question had to do, and made the final call on wording; the drafting and the sequencing suggestions were AI-assisted. The same is true of parts of the analysis in AL-I[n], where I used it to compare five responses against each other systematically rather than by impression. I am stating this in the same place I state it for the code and the architecture summaries (AL-S1, AL-S2, AL-S3, AL-I3), because an instrument is a research artefact and its authorship is a fact about the research.

Four things wrong with it, recorded rather than tidied.

1. There are three instruments, not one. P1 answered an English interview template built for a face-to-face session — it has researcher observation and reflection fields to be filled during and after. P2–P4 answered a longer Chinese self-completion questionnaire. P5 answered no form at all; the response came back as plain prose. The instrument changed between mid-June and early July and I did not re-run the earlier participant on the later form. Anything I claim across all five has to survive that.

2. Q5 primes. Read it again: *blank like a newborn, senses the real world, no language, slowly grows its own personality, each one different.* That is my project described in its own most appealing terms. Two respondents independently answered "fragile", which I find striking — but I cannot rule out that the wording produced it. A neutral version would describe the same system without "newborn" and without "grows its own personality", and I did not write one.

3. Consent and identification are not clean. P1's date, location and consent mode are blank on the returned form. P2 used a code but signed the form with a real name. P5's response has no consent block and no date at all, because it did not come back on the form. I have anonymised all five here, but the underlying records are inconsistent and I am not going to describe them as if they were not.

4. Recruitment was convenience, not sampling. Five people I could reach. Four of the five keep an animal or plant they chose; the fifth was assigned hers by a school project. That fifth case turned out to be the most useful one in the set, which tells me the sample was lucky rather than designed.

DATE

2026.6 – 2026.7

ID

AL-P0

Status

SUPPORTING

0

ID

AL-P0

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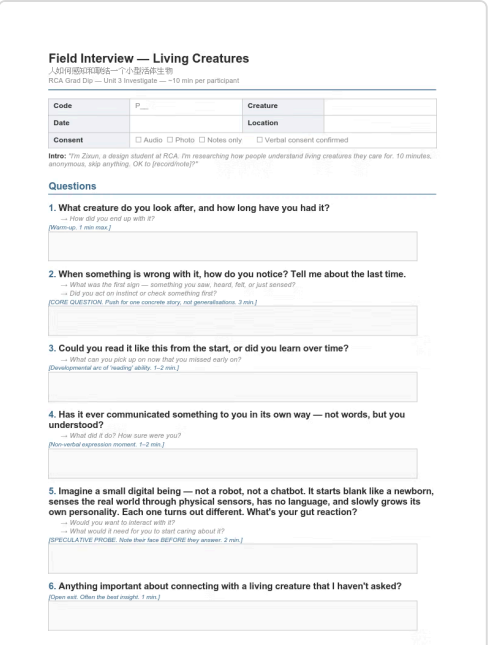

Padlet Drive

field_interview_compact.docx

FIELD INVESTGATE - SUWEN YUAN

ID

AL-P1

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suwen_yuan.wps

FIELD INVESTGATE - YUYING ZHANG

ID

AL-P2

0

田野调研问卷
你如何“读”一个活体生命
AI Life · RCA Graduate Diploma · Unit 3 Investigate
感谢你参与这项田野调研。它是设计研究项目 AI Life 的一部分，探索人如何感知并联结一个小型的活体生命。填写大约需要 10-15 分钟。你的回答会被匿名使用，仅用于本课程的研究与作品集；你可以随时停止或跳过。
请凭你自己真实的经历来写。如果某道题你确实没有相关经历，直接写“没有”或跳过即可——真实比完整重要。
关于你和你照顾的生物
你的代号（用昵称或任意代称即可，用于匿名）：
Ripple
你照顾的活体生物是（例如：某种植物 / 宠物 / 鱼 / 其他）：
一盆薄荷植物 / 我叫它“小绿灯”
你照顾它多久了：
大约三个月
正式问题
1 | 你在照顾什么活的生物？可以简单介绍一下它。
是什么、你怎么称呼它、它在你生活里大概是什么位置。
我照顾的是一盆放在窗边的薄荷。我叫它“小绿灯”，因为它不是那种很夸张、很戏剧化的植物，但每次长出新叶子时，都会像桌面上亮起一盏很小的绿色信号灯。
它在我生活里的位置有点像一个不会说话的室友：不打扰我，但会让我意识到房间并不是只属于人类。它的存在会提醒我开窗、看光线、摸一摸土，也会让我从自己的事情里短暂抽离出来。对我来说，它不是装饰品，更像是一个需要被持续读取的小生命。
2 | 请回忆最近一次你觉得它“不太对劲”的时候。你是怎么最先察觉的？
尽量具体——第一个引起你注意的细节是什么？是你看到的、听到的、闻到 / 摸到的，还是“就是感觉到了”？你当时是凭直觉，还是先去查了资料？把它当成一个小故事讲出来。
最近一次觉得它“不太对劲”，是有一天早上我发现它的叶子没有像平时那样往窗户方向微微张开，而是有点向下塌，像一把被雨打湿的小伞。最先让我注意到的不是颜色，而是姿态：它平时很“精神”，叶片边缘会轻轻翘起来，那天却像失去了支撑。
我先用手摸了土，发现表面不算特别干，但盆底附近有一点潮，味道也有点闷。我一开始以为它缺水，差点继续浇水，后来查了资料才意识到可能是通风不好加上水分太多。于是我把它移到更通风的位置，松了一点表层土，没有再急着“救它”。过了一天，新叶慢慢抬起来。我觉得那一刻挺有意思：我以为照顾就是马上行动，但它好像在告诉我，有时候不打扰也是一种照顾。

Padlet Drive

yuying_zhang.docx

FIELD INVESTGATE - QIQI

ID

AL-P3

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qiqi.docx

FIELD INVESTIGATE - Z

ID

AL-P4

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Z.docx

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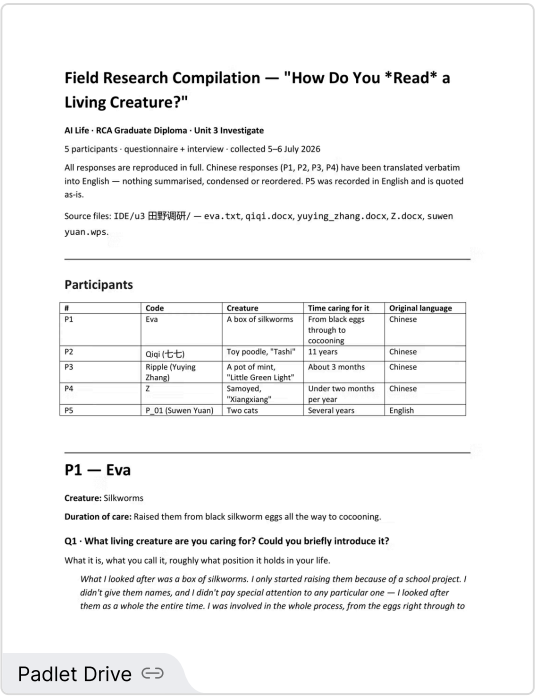

Field Research Compilation — "How Do You *Read* a Living Creature?"

AI Life · RCA Graduate Diploma · Unit 3 Investigate

5 participants · questionnaire + interview · collected 5–6 July 2026

All responses are reproduced in full. Chinese responses (P1, P2, P3, P4) have been translated verbatim into English — nothing summarised, condensed or reordered. P5 was recorded in English and is quoted as-is.

Source files: IDE/u3 田野调研/ — eva.txt, qiqi.docx, yuying_zhang.docx, Z.docx, suwen yuan.wps.

Participants

#	Code	Creature	Time caring for it	Original language
P1	Eva	A box of silkworms	From black eggs through to cocooning	Chinese
P2	Qiqi (七七)	Toy poodle, "Tashi"	11 years	Chinese
P3	Ripple (Yuying Zhang)	A pot of mint, "Little Green Light"	About 3 months	Chinese
P4	Z	Samoyed, "Xiangxiang"	Under two months per year	Chinese
P5	P_01 (Suwen Yuan)	Two cats	Several years	English

P1 — Eva

Creature: Silkworms

Duration of care: Raised them from black silkworm eggs all the way to cocooning.

Q1 · What living creature are you caring for? Could you briefly introduce it?

What it is, what you call it, roughly what position it holds in your life.

What I looked after was a box of silkworms. I only started raising them because of a school project. I didn't give them names, and I didn't pay special attention to any particular one — I looked after them as a whole the entire time. I was involved in the whole process, from the eggs right through to

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U3_Field_Research_5_Participants_EN

What five keepers actually told me — including the one who told me it doesn't work

Five respondents: two cats (P1), a mint plant (P2), a SamoyID she sees two months a year (P3), a teddy dog of eleven years (P4), and a box of silkworms raised for a school project (P5). P1 contributes only Q1–Q4; his Q5 and Q6 came back as unconfirmed placeholders and are excluded from everything below.

0

Finding 1 — nobody reads a value. Everybody reads a deviation from a baseline they built themselves.

This is the strongest thing in the set and it is unanimous among those who could read at all.

P4 did not notice her dog falling. She noticed that the sound of the landing was wrong — a dull thud instead of the light tap it had always been. P2 did not notice colour, which is what a monitoring device would report; she noticed posture — the leaves had not opened toward the window the way they usually did. P1 reads through routine: appetite and activity dropping relative to how they normally are. P3 noticed his dog pressed against him and shaking — abnormal proximity.

None of these are readable without a personal baseline. "Leaves are drooping" is not a signal; "the leaves are not doing what these leaves do" is.

What this does to the design. It is direct primary-research support for the auto-calibration argument in RE-10 — the claim that a device which learns what normal looks like for this pot, on this windowsill, in this season is doing something a mass-produced instrument with generic category thresholds cannot. I had made that argument from first principles. Four independent keepers describe exactly that mechanism as the thing they actually use.

ID

AL-I

Status

LOAD-BEARING

0

Finding 2 — the ability is learned, and it is learned by being wrong.

P4: half a year of being what she calls a "translation illiterate" — she thought whining meant the toilet, it meant snacks; she thought lying still meant illness, it meant laziness. She describes the process explicitly as a loop: guess

wrong → observe → guess again, and summarises the whole arc as going from "what I assumed" to "what I see."

P2, independently, on a plant: it never explains itself, so all she can do is keep correcting her own misreadings. She calls it learning a slow language.

P1: it took time; at the start only the obvious things.

What this does to the design. My emotion core is driven by a prediction engine — violation of expectation is the learning signal (AL-M5). I chose that from Barrett and the predictive-coding literature. Three keepers, unprompted, described their own learning in exactly those terms. That is not proof the architecture is right, but it is the first evidence I have that the mechanism I borrowed from neuroscience is also what the human on the other side of the device is doing. Both ends of this system learn by mispredicting. I had not seen that symmetry before running these.

ID
AL-I

Status
LOAD-BEARING

0

Finding 3 — the disconfirming case, and why it is the most useful response I got.

P5 raised silkworms from egg to cocoon. She reports, plainly, that she never learned to judge their state, that they never expressed anything to her, and that she formed no bond. She mostly checked whether they were still alive. When one stopped moving she did not know what had happened; she only realised it had died when all the others resumed eating and that one did not.

It would be easy to file this as a failed interview. It is the opposite. Her failure explains the other four.

Every respondent who could read their creature was reading one named individual: 扎西, 小绿灯, 香香, two cats known apart. P5 says in her own words that she cared for them as a group, gave no names, and singled none out. No individuation, therefore no baseline, therefore no deviation to read. She did not lack attentiveness — she completed the whole lifecycle. She lacked the unit of comparison.

The second difference: her relationship was assigned, not chosen. She frames the silkworms as an observation subject and part of a coursework deliverable. P2, at three months — by far the shortest voluntary tenure — had already developed fine-grained reading, because she arrived treating the plant as, in her phrase, a small life that has to be continuously read. Duration is not the variable. The stance you start with is.

What this does to the design, and it is uncomfortable. My device cannot manufacture that stance. It can support a person who already reads, and it will produce nothing for a person who does not individuate the thing in front of them. If P5 were the user, a genome-configured ecological translator would be a box that occasionally lights up. I do not currently have an answer to this, and I would rather have it stated as an open limit than quietly designed around.

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AL-I

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LOAD-BEARING

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Finding 4 — the speculative probe: two convergences and one warning.

Of the four usable responses to Q5, two produced the same first word, independently: fragile. P4 explains it as the consequence of three properties held together — starting blank, sensing through real sensors, and growing a personality — which she reads as meaning it is necessarily unrepeatable and therefore necessarily easy to damage. Her closing line is the sharpest thing anyone said to me across all five: if you let it grow into its own shape, stop asking it for the standard answer.

P2 landed on the same word plus a neighbour metaphor, and then said the thing that should worry me: if it has no language, are we not going to project onto it again — we think we have read it, and what we have actually read is our own projection.

That is my own AL-I6 argument, aimed at the reader instead of at me. In AL-I6 I established that I had authored the nurtured type's conclusion in advance. P2 is pointing out that the same failure is available to anyone standing in front of the finished object, and no amount of architectural honesty on my side prevents it. This is the single most useful criticism the field research produced, and I did not ask for it — it came out of Q6-adjacent territory in an answer to Q5.

P3 did not evaluate at all. He produced an image: a small eyed thing with no body bouncing off the table and running around, learning by touching more things. Worth recording because it is the only response that imagined the entity moving rather than being read, which is the branch I closed in RE-06.

P5's reaction was neither warm nor cold: strange. She could not decide whether such a thing counted as life or as a program, and declined to say whether she would care for one without experiencing it first. Given Finding 3, that is the most internally consistent answer in the set.

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AL-I

Status

LOAD-BEARING

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What I am not claiming.

n=5, convenience-recruited, across three different instruments, with a leading Q5 (all documented in AL-P0). Findings 1 and 2 are convergences across independent respondents and I will lean on them. Finding 3 rests on a single case and I am treating it as a hypothesis about individuation, not a result. Nothing here was collected before the architecture was designed, so none of it can be called a requirement that drove the build — it is corroboration found afterwards, and I would rather label it that way than imply a research sequence that did not happen.

ID

AL-I

Status

LOAD-BEARING

0

🥚DEEPER RESEARCH - MACHINE LEARNING🥚

🥚PLAN B

DIRECTION1--ALIFE

0

Colour	Value	Meaning
Blue	QUESTION	Poses or advances a question.
Yellow	INSIGHT	Reasoning / reflection / changing my mind — the strongest kind; don't bury it under MAKING.
Orange	PRIMARY	First-hand research (field interviews).
Purple	PRECEDENT	Literature / precedent / exhibition research.
Green	MAKING	Hands-on building / technical implementation / code files.
Grey	CONTEXT	Background setup ('what is Bazi', 'what is Silly Tavern').

SUMMARY

IDEA by Zixun

Summary by claude ops 4.6

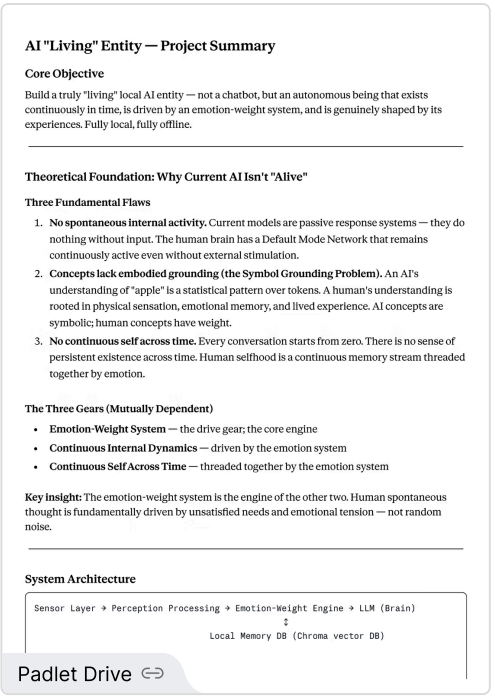
ID

AL-S1

Status

SUPPORTING

0



AI "Living" Entity — Project Summary

Core Objective

Build a truly "living" local AI entity — not a chatbot, but an autonomous being that exists continuously in time, is driven by an emotion-weight system, and is genuinely shaped by its experiences. Fully local, fully offline.

Theoretical Foundation: Why Current AI Isn't "Alive"

Three Fundamental Flaws

1. **No spontaneous internal activity.** Current models are passive response systems — they do nothing without input. The human brain has a Default Mode Network that remains continuously active even without external stimulation.
2. **Concepts lack embodied grounding (the Symbol Grounding Problem).** An AI's understanding of "apple" is a statistical pattern over tokens. A human's understanding is rooted in physical sensation, emotional memory, and lived experience. AI concepts are symbolic; human concepts have weight.
3. **No continuous self across time.** Every conversation starts from zero. There is no sense of persistent existence across time. Human selfhood is a continuous memory stream threaded together by emotion.

The Three Gears (Mutually Dependent)

- **Emotion-Weight System** — the drive gear; the core engine
- **Continuous Internal Dynamics** — driven by the emotion system
- **Continuous Self Across Time** — threaded together by the emotion system

Key insight: The emotion-weight system is the engine of the other two. Human spontaneous thought is fundamentally driven by unsatisfied needs and emotional tension — not random noise.

System Architecture

```
Sensor Layer → Perception Processing → Emotion-Weight Engine → LLM (Brain)
                                                ↕
                                    Local Memory DB (Chroma vector DB)
```

Padlet Drive

AGENT_SUMMARY.pdf

Can you keep an AI "alive" on a local machine, like a pet?

This started as a stupid question, not a research one:
could I keep an AI the way you keep a pet — living on its own machine, offline, just *there*?

I checked the technical side first and it was boring: a local language model
plus a vector memory database does it. It talks, it remembers, like **openclaw.**
But having the answer didn't settle the question, so I asked myself what I had actually described.
Two things: it *persists* (stays around between sessions) and it can be *queried* (recall). I then asked what a pet does that this doesn't — and the obvious thing is that a pet does stuff when I'm *not* interacting with it. It has states I don't trigger. My build had neither. So the check didn't answer my question, it sharpened it: the gap is between "**it persists and answers me**" and "**it has states and acts on its own**." Everything after this card is me trying to pin down what's in that gap.

ID

AL-Q1

Status

LOAD-BEARING

0

Why doesn't an AI think when no one is talking to it?

I had two separate-looking complaints and I want to show why they turned out to be one.

First: with no input, the model is idle — the weights just sit there. The brain isn't like that; even idle, the Default Mode Network keeps firing (*Raichle et al.*, 2001). So one is self-driven, the other is input-driven.
Second: and this part I didn't come up with, I ran into it.
The idea that a text-trained model never really grasps what its words *mean* is old: *Stevan Harnad* called it the ***symbol grounding problem*** in 1990, and *Bender* and *Koller* dramatised it for modern language models in 2020 with their "**octopus**" — an octopus tapping an undersea cable can learn to predict a conversation about coconuts perfectly, yet has never seen one and has no idea what the word refers to.

The moment I read it I agreed, and I kept turning it over with a more everyday example: when I say "apple" I have taste, weight, a childhood afternoon attached; the model only has the statistics of where "apple" sits in text. These two looked unrelated, so here is the step that joins them. I asked what the **DMN** is actually *doing* when it idles — replaying and recombining lived, grounded experience — and then asked what the model would idle *on*. It has nothing: its "apple" has no taste to resurface, no afternoon attached. So the reason it doesn't think unprompted and the reason its concepts are thin turn out to be the *same* reason: there's

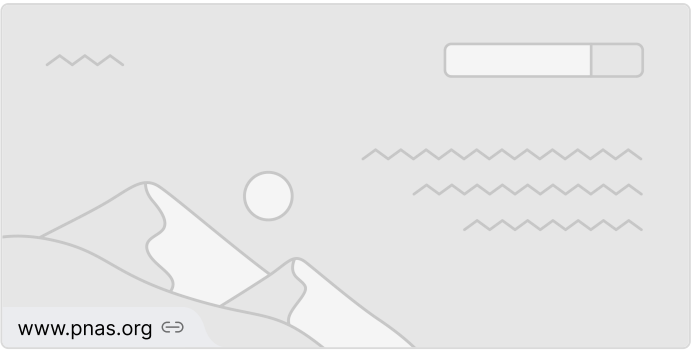
no grounded, lived material for an idle process to run on. Both point back to one absence — it isn't living in time, with a body. That's the conclusion, reached by joining the two complaints, not assumed.

ID
AL-Q2

Status
LOAD-BEARING

0

ANSWER 1 | DMN
https://www.pnas.org/doi/10.1073/pnas.98.2.676



Raichle, M. E., MacLeod, A. M., Snyder, A. Z., Powers, W. J., Gusnard, D. A., & Shulman, G. L. (2001). A default mode of brain function. PNAS, 98(2), 676–682.

ID
AL-C1

Status
SUPPORTING

0

ANSWER2 | Symbol Grounding
Harnad, S. (1990). The Symbol Grounding Problem:
https://eprints.soton.ac.uk/250382/1/symgro.pdf
DOI: https://doi.org/10.1016/0167-2789(90)90087-6

Bender, E. M. & Koller, A. (2020). Climbing:
https://aclanthology.org/2020.acl-main.463/

ID
AL-C2

Status
SUPPORTING

0

What if I wire it up to sensors?
If Q2's problem is "no grounded material," the obvious fix is a body: feed sensor streams (light, sound, temperature, motion) in as input. I want to be precise about what that fixes and what it doesn't, so I tested the idea against the three things Q2 left open.
It does fix one: grounding. "Apple" can now sit on real weight/colour/sound data, not only text — that's genuine.
But checking the other two, both survive:

1. ***Sampling vs perception*** — a sensor stream is still data *arriving*; the model still only runs when something is fed to it. The passivity from Q2 is untouched.
2. ***Emotional weight*** — every reading comes in equally weighted. Nothing gets flagged as mattering, so there's still no reason for the system to dwell on one thing over another.
3. ***Time*** — sensors add a spatial channel; they don't add continuous existence.

So the result isn't "sensors solve it" or "sensors are useless" — it's that sensors *move* the problem. They close the grounding gap and leave two walls standing, and those two are about *drive* and *time*, not perception. That tells me where to look next.

ID
AL-Q3

Status
LOAD-BEARING

0

Are the three problems separate, or does one cause the others?

Three walls left — **passivity, no emotional weight, no continuity.**

Before building three subsystems I wanted to know whether I actually had three problems or one. I worked it out in two steps.

Step 1: on passivity.

I looked again at the brain's idle activity and noticed it isn't random. When you daydream you don't think about nothing — you chew on the thing you're anxious about or the thing you want. So idle activity is directed, and what directs it is unresolved needs and fears. The consequence: just adding an internal loop won't fix passivity. A loop with nothing pulling it would only wander. It has to be pulled by something — which is exactly emotional weight, wall 2.

Step 2: on continuity.

what makes a stream of memories feel like one continuous self? The thread is that certain memories keep resurfacing while others fade, and what does that selecting is how emotionally charged they are. So continuity also rests on wall 2.

Both other walls lean on the same one. That's the conclusion I can now stand behind: the three aren't equal — **emotional weight is the driver.** So I don't need three systems, I need one engine: **a small set of hard-coded base tensions (toward connection, curiosity, stability; away from isolation, contradiction, threat) that can never be fully resolved.** That permanent unsatisfaction is what a loop can actually run on.

ID
AL-Q4

Status
LOAD-BEARING

0

What would the smallest system that runs all three actually look like?

Here the thinking has to become a build. Working back from Q4 (one driver, two driven), I tried to justify each part rather than just list an architecture:

Sensory layer: grounds concepts (the one thing Q3 did solve).
Emotion-weight engine — the driver established in Q4; without it the rest just idles.
Memory database: stamps each memory with the emotional state it formed in — because Q4 said continuity needs salience, not recency, to decide what resurfaces. A plain timestamped log wouldn't give that.
A continuous internal loop(with no external input): pulled toward whichever drive is most tense — this is wall 1 (passivity) solved through wall 2, exactly as Step 1 of Q4 demanded.
A small local language model: as the voice — deliberately small; the intelligence I care about is meant to come from the loop and the history, not from a big pretrained brain.

The one non-obvious choice is that the loop never stops. And it gives me a concrete test: when I finally talk to it, **does it answer from blank, or from whatever it was already "chewing on"?**

If the loop is doing its job, it's the latter — and that's the first observable thing separating this from a chatbot.

ID
AL-Q5

Status
LOAD-BEARING

0

When the model "remembers," what has actually changed?

The system had memory via a vector database, but I wanted to test a suspicion: is anything in the model actually changing, or am I just feeding its own past back to it through the prompt every time?

I reasoned about what a vector DB can and can't do. It stores and returns "what happened" — precise, addressable recall. But retrieving a memory and pasting it into the prompt doesn't change how the model interprets it; the

model reads tomorrow's input with exactly the same weights it used yesterday. So prompt-based memory changes what's in front of the model and never the model itself. By definition that's reminding, not changing.

For experience to change how it reads the world, the weights have to move — and locally I can do that with periodic LoRA fine-tuning on the high-salience memories. So the answer to "what changed" is a split, and I can say why each half is needed because the other can't cover it: the vector DB gives episodic recall ("what happened"); fine-tuning gives the slow reshaping of disposition ("how it now reads things"). What I'm chasing falls straight out of this: same base model + same drives + different experience should diverge at the level of disposition, not just stored facts.

ID
AL-Q6

Status
LOAD-BEARING

0

From Text to Life (Nisioti et al., 2024, arXiv:2407.09502)

ID
AL-R0

Status
SUPPORTING

0

Why I went looking for this paper at all

By the time I picked up this paper I had a working build but a nagging doubt: **was I just engineering a chatbot with extra steps, or was I actually touching a real research question?**

I went looking for whether "an AI that develops by living rather than by being pre-trained" is a recognised problem or just my private obsession.
This paper — a 2024 survey from the ITU Copenhagen ALife group — was the first thing that told me the question already has a field around it: Artificial Life. That reframing is the first thing it changed.
My project stopped being "a cool gadget" in my head and became "an entry into an existing conversation.

origin: AL-R0

ID
AL-R1

Status
LOAD-BEARING

0

The paper gave me a definition I could stand on

The paper lays out two classic ways the field defines life.
The evolutionary view (neo-Darwinism): if a system has reproduction, variation and heredity, it counts — its examples are Tierra and Avida, purely digital self-replicating creatures. The ecological view (autopoiesis, from Varela): life isn't about copying, it's about self-maintenance — a structure that keeps holding its own organisation together against a disturbing environment; its example is cellular automata. And it names the field's two goals: understanding life-as-it-is, and exploring life-as-it-could-be. What changed for me: I could finally locate my own project precisely.

AI Life isn't trying to replicate (the evolutionary view), it's trying to self-maintain a coherent internal state over time. That's the autopoiesis branch, aimed at life-as-it-could-be. Before this I had no vocabulary for that distinction; now I have an anchor I can cite.

origin：AL-R0

ID

AL-R2

Status
LOAD-BEARING

0

The mirror I didn't want to look into: does an LLM have internal states?
The most uncomfortable section is where the paper goes property-by-property through ALife's criteria and asks whether LLMs have them.

On internal states it's blunt: biological behaviour is driven by persistent internal state — developmental history, mood, physiology — and an LLM has none; its "state" is entirely the current context window, and when the window ends, the state is gone. It calls this a fundamental bottleneck, and says retrieval/external-memory tricks route around it rather than solve it. This stung because it's exactly the

gap my whole design is built to attack: Chroma for persistent memory, the emotion-weight engine for a state that carries forward, **LoRA** for accumulation in the weights. So the paper didn't hand me an idea — it certified that the hole I'd been filling is a real, named hole. What changed: I stopped treating my persistence mechanisms as implementation detail and started treating them as the actual research contribution.

ORIGIN: AL-R0
RELATED: AL-R1

ID
AL-R3

Status
LOAD-BEARING

0

R4)ELM: the first algorithm that changed how I think about "improvement"
The paper points to **ELM (Evolution through Large Models)** as one concrete way LLMs feed back into ALife.

A traditional genetic algorithm mutates randomly — it knows nothing, so it guesses and throws away ~99% of garbage. **ELM** replaces the random mutation operator with an LLM: you hand it a piece of code (like a robot's locomotion program) and ask it to modify it, and because the model has seen millions of code edits, its "mutation" is a knowledge-grounded guess, not a blind one. The result is far fewer dead offspring and even zero-shot working designs.

What changed for me was a precise correction of my own loose phrasing. I'd been thinking of **LoRA fine-tuning** as "the AI improving itself." After **ELM** I can say it sharper: improvement here is replacing uniform random sampling with knowledge-conditioned sampling. That one phrase reorganised how I describe my own system.

ID
AL-R4

Status
SUPPORTING

0

R5)LMX: the same move, made domain-independent
LMX (Language Model Crossover) is the companion technique. Traditional crossover cuts two parents at a random point and splices halves — for code, that almost always produces a syntax-error mess. **LMX** instead feeds both parents to an LLM and **lets in-context learning fuse them at the semantic level into a new offspring** — and the paper stresses it's domain-independent: code, text, math, recipe parameters, all work, because in-context learning is general.

Pairing R4 and R5 gave me the clean pair I'd been missing:

ELM swaps the mutation operator,
LMX swaps the crossover operator;
ELM needs fine-tuning on code,
LMX runs on a frozen pretrained model.

What changed: I now have two precise reference points, not one fuzzy "AI evolution" idea, and they let me describe my own design choices by contrast rather than by analogy.

ID
AL-R5

Status
SUPPORTING

0

R6)The correction that mattered most: learning lives in the population, not the weights
This is where my reading shifted from "understanding the paper" to "the paper changing a belief I held." **I'd assumed the cleverness of ELM was that the LLM gets smarter over the run.**

It doesn't. The LLM is parameter-frozen and stateless across calls — it's the same model every time. The learning is stored in the population structure (which individuals survive and breed), not in the model's weights. That is structurally the opposite of a self-learning LLM, where feedback enters the weights directly and brings model-collapse risk.

Realising this forced me to be honest about my own LoRA loop: **it is the self-learning kind, the riskier kind, and I shouldn't blur the two**. The paper handed me the contrast that lets me state the actual trade-off in my design instead of hiding it — which is exactly the kind of "approaching understanding, not a conclusion" stance the assessment rewards.

ID
AL-R6

Status
LOAD-BEARING

0

R7)Where it left me: a sharper question, and one I had to throw away
Two things changed at the end.

1. a hypothesis I proposed and the paper helped me sharpen: combine both operators (dual evolution), and add a frozen evaluator LLM as a stabiliser — which turns out to be structurally the same as the reward model in RLHF. That's a connection I now own, not borrowed.
2. the paper also closed a door honestly: on self-organisation it states plainly that current LLM agents do not exhibit it, and on autonomy that true self-sustainability is "almost out of reach" for digital agents.

Rather than ignore that, I'm keeping it visible — it marks the exact edge where my project is reaching past what the field currently claims is possible, which is a more defensible position than pretending the edge isn't there.
Net effect of the whole read: my project moved from **"a build I'm proud of"** to **"a positioned attempt inside a named field, with a stated trade-off and a known frontier."**

ORIGIN : AL-R0
ANSWERED : AL-Q6

ID
AL-R7

Status
LOAD-BEARING

0

Does it develop behaviour that was never designed into it?

This is the question everything narrowed toward, and the one I'm running rather than claiming to answer. The first problem was that my instinct was to ask "is it conscious" — and **I had to admit I can't observe that, so as a research question it's dead.**

The reframe is the key step: I borrow the criterion used in Artificial Life. Drop inner experience entirely and ask only about behaviour — do past experiences leave observable biases in future behaviour that were not in the design and not in the base model? That version is measurable.
Then, to read it cleanly, **I need something to measure divergence against. That's why I run two isolated tracks: a "wild type" with no language and no caregiver, randomly initialised; and a "nurtured type" raised by a caregiver model in a language-rich world.**
The logic of the comparison is the whole point — anything that appears in one track and not the other cannot be explained by the shared base model, so it has to come from the environment and the history. Whatever survives that test is the first evidence the system grew something it was never given. The rest of this Padlet is that experiment.

this is a two-arm controlled comparison, borrowed from experimental biology — same base architecture, one variable changed, and the finding lives in the *difference* rather than in either arm on its own. I use it because I cannot observe an internal state directly. A control is the only thing that lets me attribute a behaviour to history rather than to the base model.

ID

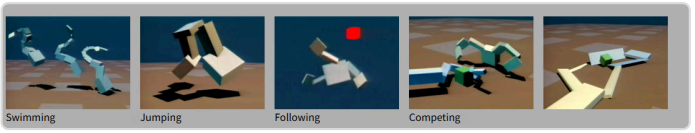
AL-Q7

Status

LOAD-BEARING

0

Karl Sims, Evolved Virtual Creatures (1994) — the criterion I needed, written thirty years earlier
https://www.karlsims.com/evolved-virtual-creatures.html
Karl Sims, Evolved Virtual Creatures (1994)



Paper: Sims, K. (1994). "Evolving Virtual Creatures." SIGGRAPH '94 Proceedings, 15–22.
Studied from documentation and the original SIGGRAPH paper; I have not seen the work installed.
Sims built a population of virtual block creatures inside a simulated physics world, each one's body *and* its neural wiring described by the same directed-graph "genetic language". Creatures were scored on a task — swimming, walking, jumping, following, competing for a cube — and the successful ones had their graphs copied, combined and mutated into the next generation. Run the loop long enough and locomotion strategies appear that nobody wrote.
What it gave me is a sentence, not an idea. His paper says the successful strategies that emerge are ones that would be difficult to invent or build by design. That is exactly the criterion AL-Q7 is groping for. My version of the question — *do past experiences leave observable biases in future behaviour that were not in the design and not in the base model* — is the same test with the population swapped out for a single lifetime. Knowing the field had already stated it this cleanly in 1994 stopped me phrasing it as though I were inventing the question.
Where I go the other way, and why it matters. Sims's learning lives in the population: which individuals survive and breed. The model of each creature is not getting cleverer — the *distribution* is. This is the same structural point I had already reached in AL-R6 about ELM, where the LLM is frozen and the learning sits in the population. My system does the opposite. One

individual, no reproduction, no selection, and the accumulation goes into the weights via LoRA. That is the riskier of the two architectures and I said so in AL-R6. So Sims is not an influence on this build. He is the clearest available example of **the branch I did not take**, and having a name and a date attached to that branch is more useful to me than another paper agreeing with me.

ID
AL-C6

Status
SUPPORTING

0

Špela Petrič, PL'AI (2020) — someone else hit the instrument problem, and said so
Špela Petrič, PL'AI (2020)
https://www.spelapetric.org/plai

Residency notes (the instrumentation passage): https://www.spelapetric.org/digitaldeal-residency

V2_ Lab for the Unstable Media:
https://v2.nl/works/plai/

Studied from the artist's project documentation and residency writing; I have not seen the work installed.
PL'AI is a three-metre robot built to play with cucumber plants. Thirty-six reels lower bouncy balls on steel wires into a planter below; a camera watches for the moment a tendril grabs a ball, and a 3D scanner sweeps the plants every fifteen minutes. A neural network turns the scans into an abstract matrix of the plant and samples it to decide where to send the wires next. Over months the wires become a trellis and the plants fruit through it. The premise is that play is a condition of being alive, including for plants, and that a machine can be built to enter plant time rather than to extract yield from it.
The reason this card exists is one sentence in her residency writing, not the robot. She describes the sensors as limited in their modalities and capacities, and says the team therefore had to build a set of tools purely to visualise what was happening inside the machine's decision-making — hybrid video-and-simulation representations, made in order to see what the robot was actually sensing. That is symbol_tracker.py. That is the same problem, arrived at independently, by someone with a research team and a budget. It tells me the observation instrument is not a weakness in my setup that a more competent person would have avoided — **it is structural to this kind of work.** If your subject cannot speak, you have to build a second thing whose entire job is to look inside the first thing, and that second thing is now also yours to be wrong about.
Where I can say something back rather than only agree. I built mine before the run (AL-M32) and then audited it (AL-I17), and the audit found that my score was awarding its highest marks to a system that had stopped moving — one cluster, stability 1.0, state 6 → state 6 → state 6, because a frozen system is perfectly predictable. I do not know whether PL'AI's team interrogated their visualisation tools the same way; the documentation I can reach does not say. What I can say is that **the tool needs a stated null case or it cannot fail**, and until a measure can fail, a good reading from it means nothing. That requirement is the one thing I have that this precedent does not hand me.
One smaller thing, on presentation. Visitors to PL'AI cannot watch play happen — it runs at plant speed. What they get is a screen replaying the last twenty-four hours as time-lapse. That is the same problem AL-C4 raised through Richard Long: a process that takes longer than an attention span has to be compressed into something readable at a glance, and the compression is authored.

ID
AL-C7

Status
SUPPORTING

0

Madeline Gannon, Mimus (2016) — legible state without language, solved in movement instead of light
Madeline Gannon (ATONATON), Mimus (2016)
https://atonaton.com/mimus

Design Museum Q&A: https://designmuseum.org/qa-with-madeline-gannon
Shown in 'Fear and Love', Design Museum London, 2016–2017

Studied from the artist's studio documentation and the Design Museum's material; I have not seen the work installed. Mimus is a 1,200kg ABB industrial arm — the kind that normally spot-welds cars — reprogrammed to have no pre-planned movements at all. Depth sensors in the ceiling let her see everyone in the room at once. If someone interests her she comes closer and follows them for a while; if you stand still too long she tries to get your attention, then gets bored and goes to find someone else. Gannon's stated design method is to borrow from how we read animals: a dog's tail, a cat's crouch. Seen from a distance she rears up and looks down at you like a bear; up close she approaches from below like an excited dog.
Why this belongs in Investigate rather than in Realise. TC-01 is the card where I write that if the Plan A entity has no language, I have no way of knowing whether it developed any internal state — the only possible evidence would be some expressive behaviour it invents, a pattern a person could learn to read without it ever being translated into words. Mimus is that problem, taken seriously, by someone who got it working in a gallery. It is the strongest evidence I have that the bet underneath Plan A is a real design problem with a real literature, and not a thing I made up to justify a device.
The difference that is useful to me: she solved it in kinematics, I chose light. Posture, approach speed, height, hesitation — none of it needs a screen, a colour key, or a legend. AL-C3 (FUKT) is where I hold myself to the fact that an emotional state has no natural appearance and that my LED mapping is therefore an authored translation I am accountable for. Mimus sharpens that: it demonstrates the same job being done in a completely different medium, which means **light is a choice I made, not the only channel available.** That makes the accountability in AL-C3 heavier, not lighter, and it is worth writing down that I noticed.
The difference I should not paper over. Mimus's vocabulary was authored. Gannon decided what "interested" and "bored" would look like, and designed the postures so a stranger could read them in seconds — which is why it works. My whole claim in AL-Q7 is that I am looking for behaviour that was *not* designed in. So this is not a precedent I can point at and say "like that." It is closer to the opposite: **Mimus shows what a fully authored expressive vocabulary buys you — instant legibility — and therefore what I am giving up by refusing to write one.** If my creature never becomes readable, Mimus is the standing reminder that legibility was available all along and I chose not to purchase it that way.

ID
AL-C8

Status
SUPPORTING

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AI Life-Form Architecture Validation: A Neuroscience Literature Review
written and research by claude opus 4.6

ID
AL-S2

Status
SUPPORTING

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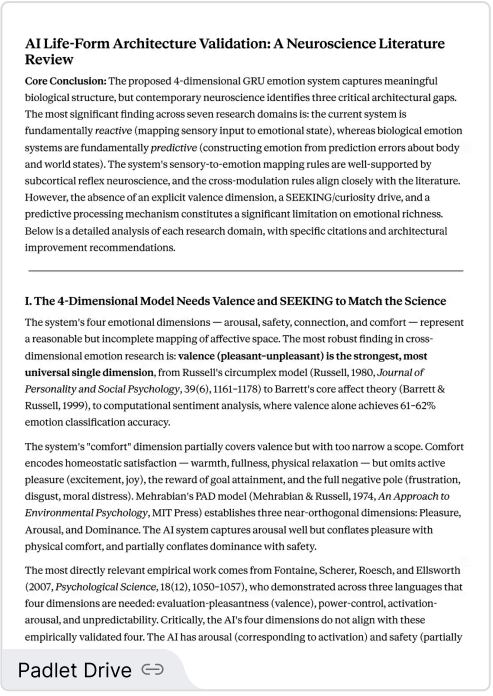

AI Life-Form Architecture Validation: A Neuroscience Literature Review

Core Conclusion: The proposed 4-dimensional GRU emotion system captures meaningful biological structure, but contemporary neuroscience identifies three critical architectural gaps. The most significant finding across seven research domains is: the current system is fundamentally *reactive* (mapping sensory input to emotional state), whereas biological emotion systems are fundamentally *predictive* (constructing emotion from prediction errors about body and world states). The system's sensory-to-emotion mapping rules are well-supported by subcortical reflex neuroscience, and the cross-modulation rules align closely with the literature. However, the absence of an explicit valence dimension, a SEEKING/curiosity drive, and a predictive processing mechanism constitutes a significant limitation on emotional richness. Below is a detailed analysis of each research domain, with specific citations and architectural improvement recommendations.

I. The 4-Dimensional Model Needs Valence and SEEKING to Match the Science

The system's four emotional dimensions — arousal, safety, connection, and comfort — represent a reasonable but incomplete mapping of affective space. The most robust finding in cross-dimensional emotion research is: **valence (pleasant–unpleasant) is the strongest, most universal single dimension**, from Russell's circumplex model (Russell, 1980, *Journal of Personality and Social Psychology*, 39(6), 1161–1178) to Barrett's core affect theory (Barrett & Russell, 1999), to computational sentiment analysis, where valence alone achieves 61–62% emotion classification accuracy.

The system's "comfort" dimension partially covers valence but with too narrow a scope. Comfort encodes homeostatic satisfaction — warmth, fullness, physical relaxation — but omits active pleasure (excitement, joy), the reward of goal attainment, and the full negative pole (frustration, disgust, moral distress). Mehrabian's PAD model (Mehrabian & Russell, 1974, *An Approach to Environmental Psychology*, MIT Press) establishes three near-orthogonal dimensions: Pleasure, Arousal, and Dominance. The AI system captures arousal well but conflates pleasure with physical comfort, and partially conflates dominance with safety.

The most directly relevant empirical work comes from Fontaine, Scherer, Roesch, and Ellsworth (2007, *Psychological Science*, 18(12), 1050–1057), who demonstrated across three languages that four dimensions are needed: evaluation-pleasantness (valence), power-control, activation-arousal, and unpredictability. Critically, the AI's four dimensions do not align with these empirically validated four. The AI has arousal (corresponding to activation) and safety (partially

Padlet Drive

agent_en.pdf

Project phase devided

skills support and plan generated by claude ops 4.6

ID

AL-S3

Status

SUPPORTING

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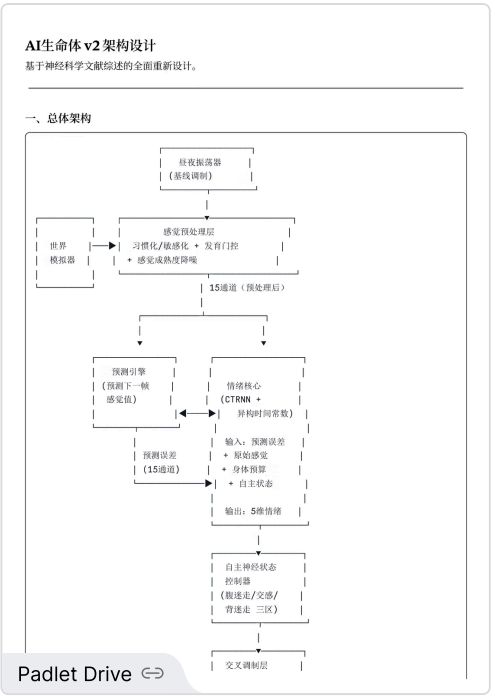

AGENT_PHASES.pdf

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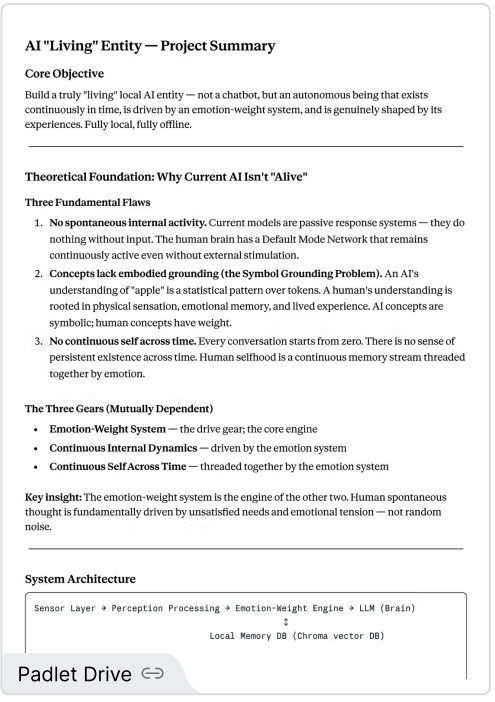

AGENT_SUMMARY.pdf

Why I replaced the GRU emotion core with CfC + AutoNCP

My first emotion engine was a **GRU**: four sensory inputs, a 64-unit hidden layer, four emotion outputs, learning online. It ran, but watching it run I kept hitting the same three limits, and each one is the reason I changed something.

1. it was purely reactive — it mapped the current reading to a current feeling, with no prediction; but the literature I was reading on idle brain activity treats spontaneous thought as a predictive process, so a reactive core couldn't ever produce the unprompted "inner life" I was after.
2. it had no real valence axis — it tracked arousal-like movement but not good-vs-bad, and without valence "this matters and it's bad" is unrepresentable.
3. the GRU's fixed time-step sat badly with the fact that my world arrives at irregular intervals. **CfC** (closed-form continuous-time networks, Hasani et al., 2022) answered the third directly: it's built for continuous, irregularly-sampled time, derived from liquid networks modelled on small biological nervous systems. AutoNCP gave me a sparse, wired neuron topology instead of a dense black box, which matters because I want to be able to point at which pathway carries which drive.

So the change wasn't chasing a better benchmark — it was a literature-driven correction: I swapped the material (the network) when the old material couldn't hold the three properties my own argument required.

What it revealed: the moment the core became predictive rather than reactive, the system's idle output stopped looking like noise and started looking like it was waiting for something.

ID

AL-I1

Status

LOAD-BEARING

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Step 1: Design Concept & Technology Selection

Status

SUPPORTING

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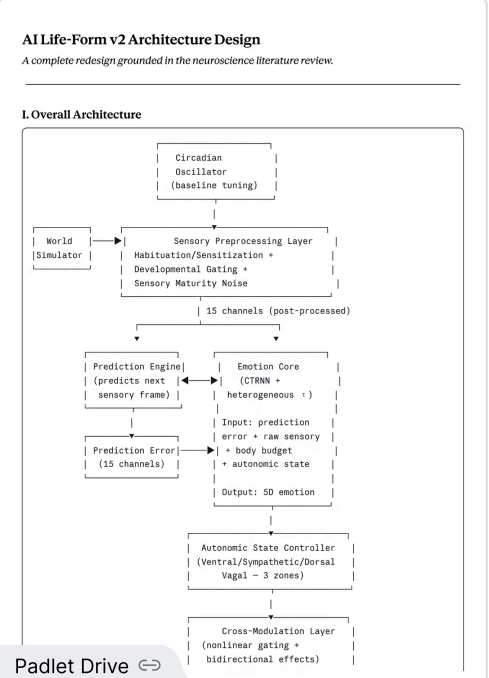

Padlet Drive

phase_en.pdf

Built a Python-based AI life simulator with genuine emotional dynamics

- **Core model:** CfC neural network, 64 neurons, 5-dimensional emotion space
- **Emotion dims:** Valence / Arousal / Safety / Connection / Seeking
- **Theory base:** Barrett predictive emotion, Polyvagal theory, Bowlby attachment

ID

AL-M1

Status

RUNNING

0

Step 2: Phase 1 — Building the Core Emotion Network

What we did in this file: Built four core modules.

- *SensoryPreprocessor*: normalises 15-channel sensory input via habituation.
- *PredictionEngine*: prediction-error engine generating violation-of-expectation signals.
- *EmotionCore*: **the CfC network trained with a triple learning rate (readout / hebbian / pred).**
- *TargetGenerator*: computes target emotions from sensory input with built-in separation dynamics (Protest 0–50 frames → Despair 50–200 → Detachment 200+).

ID

AL-M5

Status

RUNNING

0

```
#!/usr/bin/env python3
"""
life_v2.py — AI生命体v2核心 (Phase 1: 婴儿期纯数值训练)
=========================================================
世界模拟器(15通道五感) → 感觉预处理(习惯化/敏感化+发育门控)
→ 预测引擎(预测误差) → CfC情绪核心(5维) → 在线学习

婴儿期设计为连续运行五天。无语言模型。

依赖:  pip install torch ncps
运行:  python life_v2.py
"""

import time, math, random, datetime, os, signal, sys, json
import torch

# Windows GBK 终端无法显示 emoji，统一切换为 UTF-8 输出
if hasattr(sys.stdout, 'reconfigure'):
    sys.stdout.reconfigure(encoding='utf-8', errors='replace')
if hasattr(sys.stderr, 'reconfigure'):
    sys.stderr.reconfigure(encoding='utf-8', errors='replace')
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim

try:
    from ncps.torch import CfC
    from ncps.wirings import AutoNCP
except ImportError:
    缺少 ncps 库 / missing ncps library. 请运行 Run: pip install
```

Padlet Drive

file：life_v2.py

Step 3: Phase 2 — Layering Biological Systems

file：life_v2.py

What we did: Layered three physiological regulation modules.

- CircadianOscillator: modulates arousal and positive affect across the sleep/wake cycle.
- BodyBudgetSystem: tracks continuous depletion and replenishment of energy, hydration, and thermal regulation. AutonomicController: three-zone polyvagal model — zone 0 dorsal vagal (freeze), zone 1 sympathetic (fight-flight), zone 2 ventral vagal (safe/social).

Also introduced Hebbian learning and developmental pruning to naturally sparsify network connections over time. **Barrett's Theory** of Constructed Emotion argues that emotions are fundamentally the brain's predictive regulation of the body budget. Without these systems, the model is merely an abstract emotion mapper, not an embodied agent. Polyvagal Theory (Porges) is the leading neuroscientific framework for infant socio-emotional development. Hebbian pruning simulates the infant brain's transition from a hyper-connected state to efficient neural circuits.

ID

AL-M6

Status

RUNNING

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Supplementary Step: Pre-Integration Smoke Test for the World Simulator

What we did: Before wiring WorldSimulatorV2 into the main training loop in life_v2.py, wrote a standalone debug script to run the world simulator in isolation — verifying that sensory output values fell within expected ranges (light, sound, skin temperature, etc.) and that the event system could trigger correctly. Once it passed, the script was left untouched — a one-time smoke test.

Why: The world simulator is the input source for the entire training system — if it produces faulty sensory data (value overflow, events failing to trigger, missing fields), any anomaly observed after connecting it to the emotion network becomes impossible to attribute to the network or the environment. Validating the environment layer in isolation before integration narrows the

```
#!/usr/bin/env python3
"""
test_world.py — 独立调试脚本
==============================
只跑 world_v2.py，不启动 CfC 训练，用来确认:
  1. openai 包能导入 (HAS_OPENAI=True)
  2. 逆向 API (pro.chr1.com) 能响应
  3. LLM 编剧线程能成功生成并解析事件脚本
  4. EventTranslator 不再抛 TypeError

用法:
  python -u test_world.py

观察终端，看到 [编剧] ✓ 字样即为成功。
"""

import sys, time, datetime

# ── 1. 检查 openai 包 ──────────────────────────────────────
try:
    from openai import OpenAI
    print("[依赖] ✓ openai 包已安装")
except ImportError:
    print("[依赖] ✗ openai 未安装，请先 pip install openai")
    sys.exit(1)

# ── 2. 导入模拟器 ──────────────────────────────────────────
try:
    port WorldSimulatorV2, LLM_BASE_URL, LLM_MODEL, LLM_TIMEOUT,
    orld_v2 加载成功")
```

Padlet Drive

test_world.py

debugging scope to a single module, avoiding back-and-forth guessing across two coupled systems later.

ID
AL-M7

Status
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Added three physiological regulation systems to the infant agent

1. Circadian Oscillator: 24-hr PA/arousal rhythm, matures with development
2. Body Budget System: Energy / hydration / thermal — deficits modulate emotion targets
3. Autonomic Controller: Dorsal vagal (freeze) → Sympathetic (fight-flight) → Ventral vagal (safe)

ID
AL-M8

Status
RUNNING

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Added MBTI + attachment style to parents so personality drives parenting behavior

1. Mother (ISTJ): High conscientiousness, anxious-avoidant attachment, authoritarian parenting
2. Father (ISFJ): High agreeableness, permissive parenting, harmony-first, avoids confrontation

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AL-M8

Status
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Step 4: World Simulator

What we did: Provided a realistic family environment.

A rule engine generates circadian baseline sensory values. LLMPlaywright background thread calls a language model every few minutes to generate 10–25 minute event scripts. worldbook.py defines family characters (Mother Xiaoyü, Father Mingzhe, Grandma), daily schedules, and 22 event types (feeding, diaper change, singing, family argument, night waking, etc.), serving as the LLM's system prompt.

reason recall： Real development does not happen in a vacuum — an infant's emotional state is constantly shaped by caregiver behaviour, environmental noise, and bodily needs. A pure rule engine produces predictable periodic patterns, causing the model to overfit fixed sequences. The LLM playwright injects emergence and unpredictability, making each training run unique and testing the model's generalisation to open-ended contexts.

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world_v2.py

ID

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Status

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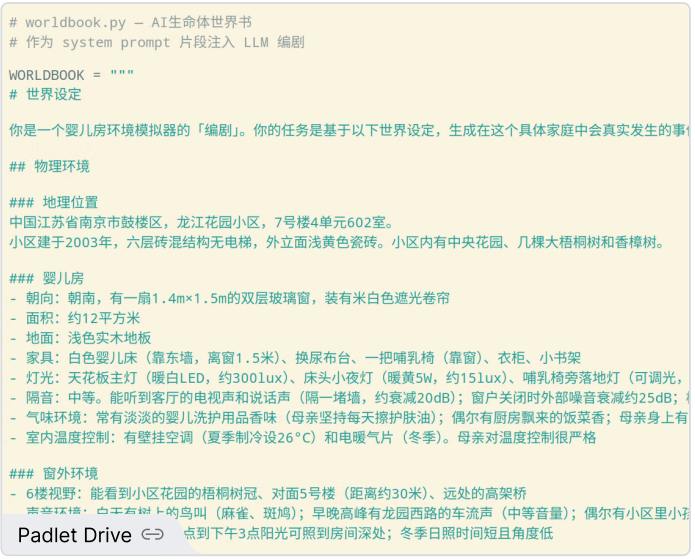

worldbook.py

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AL-M9

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worldbook_cpython_310.pyc

Step 5: Discovering & Fixing Three Training Bugs At 44.8% dev progress

file：life_v2.py

1. Bug 1 – Arousal frozen: Always 73.1, zero variance — sympathetic gating multiplied all variance to a fixed point
2. Bug 2 – Thermal floor: Sleep recovery (0.00015/frame) far below drain (0.001/frame) — never recovers
3. Bug 3 – Permanent detachment: 86.2% in sep_phase=3, no_social_count reached 5011 with no recovery mechanism

Bug 1 — Arousal Freeze：乘法更新 aro_t × 1.2 + 10 把基础值恒定映射为 73，方差趋零。修复：改为纯加法 aro_t + 20。
Multiplicative update aro_t × 1.2 + 10 mapped the baseline to a constant 73, collapsing variance to near-zero.
Fix: switched to pure additive aro_t + 20.

Bug 2 — Thermal Recovery Failure：皮肤温度传感器常态约 25°C，低于 33–37°C 恢复
窗口，体温持续亏损。修复：恢复区间拓宽到 31–38°C，睡眠时暂停消耗，固定速率恢复。
Skin temperature sensor reads ~25°C in ambient conditions, below the 33–37°C recovery window.
Fix: widened range to 31–38°C, suspended drain during sleep, used a fixed recovery rate.

Bug 3 — Permanent Detachment Lock：no_social_count 无上限无衰
减，可累积至 5,000+。修复：加 400 帧上限，弱社交信号时触发反向衰减。
no_social_count had no ceiling and no decay, accumulating to 5,000+.
Fix: added a 400-frame cap and a reverse-decay branch on weak social signals.

ID
AL-M10

Status
RUNNING

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Step 6: Phase 3 / 4 Interface Placeholders

file： life_v2.py
What we did: Inserted comment markers in the code to reserve injection points.

Phase 3： injects SocialResonanceSystem, implementing Stern (1985)
affect attunement and vagal maturity modelling; trigger: dev ≥ 50%.

Phase 4：injects ProtoIntentionalitySystem, implementing Tomasello (1995)
joint attention and violation-of-expectation surprise signals; trigger: dev ≥ 70%.

reason recall：Developmental psychology maps infant cognition into clear stages. Using placeholders rather than writing all code at once lets the model fully develop baseline emotional capacities before introducing higher-order social cognition, preventing premature complexity from interfering with foundational learning.

ID
AL-M11

Status
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Step 7: Scheduled Task — Automated Code Injection

life_v2.py、phase_status.json

What we did: Built a scheduled task (every 30 minutes) that automatically monitors developmental progress in life_log.jsonl. Once a threshold is reached, it injects Phase 3 / 4 code at the marker positions, runs a py_compile syntax check, and sends a system notification to prompt restart. Injection state is persisted in phase_status.json to prevent double-injection.

Why: Training runs at one frame every 3 seconds; reaching the Phase 3 threshold could take days. Manual monitoring is impractical — an automated mechanism is required to trigger phase upgrades at precisely the right moment, running correctly even when the user is away.

ID

AL-M12

Status

SUPPORTING

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{"phase3_applied": true, "phase4_applied": true}

Padlet Drive

phase_status.json

Step 8: Real-time Monitoring Dashboard

What we did: Built a Flask + Chart.js 4 monitoring panel. 5 status cards show current emotion values with colour coding and descriptors. Emotion trajectory chart (5D overlay with gradient fill), sensory input chart, and training loss chart displayed side by side. Narrative bar shows the LLM-generated scene description. Status bar contains age, event label, separation phase, learning rate, dev progress.

Fixed two Chart.js 4 issues: canvas height controlled via wrapper div (CSS on canvas does not work in Chart.js 4); switched fill:true to fill:false to prevent white overlay when stacking multiple datasets.

Why: without real-time visualisation, it is impossible to distinguish normal development from a bug. The dashboard was the primary tool that enabled discovery of the three structural bugs, and makes the training process presentable and interpretable to others.

ID

AL-M13

Status

SUPPORTING

0

ID
AL-M13

0

```
#!/usr/bin/env python3
"""
dashboard.py — AI生命体实时监测面板
====================================
读取 life_v2.py 写入的 life_log.jsonl，在浏览器中显示实时情绪曲线。

依赖:  pip install flask
运行:  python dashboard.py
浏览器打开:  http://localhost:5000
"""

import json, os, time
from flask import Flask, Response, jsonify

app = Flask(__name__)

# 始终在 dashboard.py 所在目录找日志，不依赖启动目录
_HERE    = os.path.dirname(os.path.abspath(__file__))
LOG_FILE = os.path.join(_HERE, "life_log.jsonl")

def read_tail(n=300):
    """读取日志文件最后n行"""
    if not os.path.exists(LOG_FILE):
        print(f"[dashboard] 日志文件不存在: {LOG_FILE}", flush=True)
        return []
    try:
        with open(LOG_FILE, 'r', encoding='utf-8', errors='replace') as f:
            lines = f.readlines()
        print(f"[dashboard] 读取成功: {len(lines)} 行", flush=True)
```

Padlet Drive

dashboard.py

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Padlet Drive

life_log.jsonl

Dashboard: Full Bilingual UIWhat changed

Every label in the monitoring dashboard (dashboard.py) now displays Chinese and English side by side.

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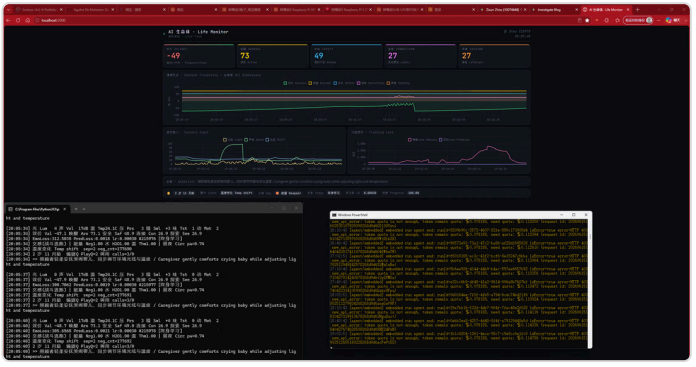

---

until 6.15 23:33

ID
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Step 9: Switching to a Local LLM via Ollama

What changed: The narrative playwright in world_v2.py was redirected from a paid cloud API (DeepSeek-V3.2 at pro.chr1.com) to a locally running model via Ollama (qwen3:14b, 9.3 GB, running 100% on local GPU).

Technical details:
LLM_BASE_URL → http://localhost:11434/v1 (Ollama's OpenAI-compatible endpoint)
LLM_MODEL → qwen3:14b

max_tokens increased from 1500 → 2000 (local model needs more token headroom before producing output)
temperature lowered from 0.85 → 0.4 (more stable JSON generation)
response_format: json_object removed (incompatible with Ollama's qwen3 — causes empty content)
options: {think: false} added via extra_body to disable qwen3's default chain-of-thought mode
Regex stripping of ... blocks added as a fallback

Why: The external API ran out of credits. Switching to a local model eliminates ongoing API cost and latency, keeps all computation on-device, and maintains full functionality. The playwright generates world events and narrative text — it is a pre-trained tool, not the model being trained. The CfC network (the actual "life agent") continues to train independently and is unaffected by this change.

ID

AL-M14

Status

RUNNING

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Developmental stage: dev=1.0, and both phase3_applied and phase4_applied are true in phase_status.json. Stage shows "终身学习" (Lifelong Learning). All developmental milestones have been completed — the agent is now in continuous learning mode and won't trigger any new "growth events." Virtual age moved from 3y0m to 3y1m, confirming the virtual clock doesn't map linearly to real training time.

Loss trend (healthy): emo_loss dropped from an average of ~498 at the start of this window to ~293 at the end (-41%), and pred_loss dropped from 0.0178 to 0.0042 (-76%). Both are converging steadily with no oscillation or stagnation — the network is still learning effectively, not idling.

Emotional trajectory (improving, but not yet positive): valence averaged -49 → -30, safety went 47 → 59, seeking went 27 → 37. All moving in a healthy direction, but valence is still negative overall — the subjective baseline is less bad, not yet good.

An anomaly worth flagging: the arousal dimension has zero variance across all 5154 records — it's frozen exactly at 73.10, never moving once. This matches the symptoms of the "arousal freeze" bug fixed in an earlier session — it's likely either recurred, or that fix only resolved that specific trigger without addressing the root cause. Worth checking the readout/hebbian update path for arousal in EmotionCore to confirm the gradient isn't getting clamped to zero, or that some learning-rate branch isn't silently failing. The connection dimension is also nearly frozen (variance is tiny, ranging only 26.9–32.2).

Separation state stuck in an unhealthy range: across the full log, only 17.8% of time is in "normal," 37% in protest, 37.7% in despair, 7.5% in detach — over 80% of the time is outside normal. Looking at the most recent 500 records, this hasn't improved — despair actually rose from 37.7% to 42.4%. The frozen connection dimension may be part of why the separation-coping mechanism can't repair itself back to normal.

Autonomic system (good news): across the most recent 500 records, auto_zone is 100% zone 2 (ventral vagal / safe). Physiological regulation is stable, with no oscillation between freeze and fight-flight.

Overall: the training is numerically healthy but emotionally it's stuck — valence stays negative, separation state spends most of its time in protest/despair, and there's a hard bug where arousal is completely frozen. I'd prioritize tracking down the arousal freeze first — without fixing it, the subjective emotional layer will likely stay stuck in its current state even as the loss numbers keep improving.

DATE
2026.6.15

ID
AL-I2

Status
RUNNING

0

Why I Made This Change

I used AI tools to help write the code and run the analysis — first having it pull the training logs and compute statistics: loss trends, variance across emotion dimensions, the distribution of separation-phase states. That let me reason from data rather than impression when judging whether training was actually going well.
I'd felt that this worldbook simulator was too monotonous. MBTI labels pinned every character to a fixed reaction pattern — the mother always behaving exactly like an ISFJ "should," the father always like an ISTJ "should," the environment reduced to a few abstract roles (nursery, parents, grandma) with none of the specific, hard-to-name texture of real life — an old scar, an heirloom spoon, a clock that runs a few minutes slow. What I wanted to fix wasn't "are there enough event types," it was "does this world feel like it actually existed."
So instead of discarding MBTI entirely, I kept it as a baseline personality and layered "state-dependent contradiction" on top of it — fatigue, mood, and hormonal cycles pull a person away from their "textbook" personality, which is much closer to how real people behave. I also gave each character unique, idiosyncratic details so they'd stop being the embodiment of a label and start being a specific person. I deliberately scoped this change to worldbook.py only, leaving the underlying event-generation logic untouched, to keep the blast radius small.

But once the enriched worldbook was running again, I noticed the generated narratives included things that don't exist in my setting at all — orange blossom scent, piano sound. I used AI tools to help me trace through the code, and found a more fundamental problem: world_v2.py never actually sent WORLDBOOK to the LLM in the first place. In other words, no matter how rich and three-dimensional I made the character settings, the model's actual prompt contained none of it — it was inventing content from nothing rather than working from the world I'd built. That's exactly why this had to be fixed: without it, everything I'd written upstream would have had zero effect on what actually got generated — like writing a detailed script that no one ever performs.

After fixing that, I found a related issue: restoring the world setting made the prompt much longer, and the "bilingual narrative" rule got buried inside it — the model stopped following it, and narratives reverted to pure Chinese. That made me realize: the longer an instruction set gets, the more the easily-overlooked critical requirements need to be pulled out and stated on their own, rather than trusting the model to extract priorities from a long block of text.

In the end, I chose to restart training from scratch, because I wanted a genuinely clean new model — not one with early, ungrounded, buggy records mixed into its real training history. Even if it was only a few minutes' worth of entries, I felt it was worth a clean restart so that, looking back at the training curve later, there wouldn't be a "fake" opening segment contaminating the data.

ID
AL-I3

Status
LOAD-BEARING

0

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step10：Discover the Missing Injection

File: world_v2.py

What was done: Discovered that WORLDBOOK (family personas, contradiction layer, environment details) was imported but never actually included in the prompt sent to the LLM.

Why: An earlier fix for the qwen3 "empty content" bug had simplified the system message, accidentally stripping out the world setting entirely — leaving the model with zero grounding, so it invented details that don't exist in the setting (orange blossom smell, piano sound).

ID

AL-M15

Status

RUNNING

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world_v2.py

0

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life_log.jsonl

0

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emotion_weights_v2.pt

Step 11： Reconnect the World Setting

File: world_v2.py → _generate() method

What was done: Concatenated WORLDBOOK content into the system message, with an added constraint not to invent people/smells/sounds outside the established setting.

Why: To keep the LLM's generated events and narratives grounded in the actual family identities, personality contradiction layer, and physical environment — instead of free-associating.

ID

AL-M16

Status

RUNNING

0

Step 12：Expand the Context Window

File: world_v2.py (same API call)

What was done: Raised Ollama's num_ctx from default to 16384.

Why: WORLDBOOK + event-type list + output schema together total roughly 6500 tokens — well beyond qwen3's default

context window. Without expanding it, the prompt would be silently truncated, possibly cutting off the start of the world setting.

ID

AL-M17

Status

SUPPORTING

0

Step 13：Reinforce the Bilingual Rule

File: world_v2.py (same system message)

What was done: After the first restart, narratives reverted to pure Chinese with no English. Added a standalone "narrative field must be bilingual" rule to the system message, with a format example.

Why: The bilingual requirement was originally only inside OUTPUT_SCHEMA, and got diluted by the much longer world setting — the model stopped following it. A standalone rule plus example made compliance consistent.

ID

AL-M18

Status

SUPPORTING

0

Step 14：Back Up and Restart Clean & Verify the Fix

File: life_log.jsonl, emotion_weights_v2.pt (on Windows)

What was done: After each code change, killed the currently running training process, renamed (not deleted) the old log/checkpoint as a backup, and started a brand-new process. Did this twice — once after the injection fix, once after the bilingual fix.

After ~4 minutes of fresh running, checked the new log: narratives are consistently bilingual and reference concrete worldbook details (the crib's wind chime, the mother's enamel scoop, dance-precise movements); 13 distinct event-type combinations already appeared in that short window.

Why: The user explicitly asked for training from scratch, and this avoided mixing ungrounded pre-fix script entries into the real training history.

To confirm the entire fix chain (grounding injection + larger context + bilingual reinforcement + clean restart) is actually working, not just superficially appearing to.

ID

AL-M19

Status

SUPPORTING

0

0

0

```
#!/usr/bin/env python3
"""
sensor_bridge.py — Arduino 物理传感器串口桥
============================================
实验A（野生型）的感知来源。

硬件接线：
  BH1750  (光照)      → I²C (SDA/SCL)
  MAX9814 (声级)      → A0 (ADC)
  DHT22   (温湿度)    → D2 (Digital)
  HC-SR501(人体红外) → D3 (Digital)
  FSR402  (压力)      → A1 (ADC)
  MPU6050 (加速度计) → I²C (SDA/SCL)

串口协议：Arduino 每 CYCLE_SEC 秒发送一行 JSON:
  {"lux":320,"db":45,"temp":24.5,"hum":60,"pir":1,"prs":512,"ax":0.1,"ay":-0.02,"

本模块支持两种模式：
  1. 真实模式 (real)  — 从USB串口读取Arduino数据
  2. 模拟模式 (sim)   — 用真实时钟+噪声生成伪传感器数据（无Arduino时调试用）

依赖：pip install pyserial (仅真实模式需要)
"""

import time, math, random, json, sys, os, datetime, threading
from collections import deque

# ═══════════════════════════════════════════════════════════════
#  空间)
```

Padlet Drive

sensor_bridge.py

0

```
#!/usr/bin/env python3
"""
symbol_tracker.py — 内部符号涌现监测器
========================================
实验A（野生型）最核心的观测工具。

功能：
  1. 持续记录CfC隐藏状态 h(t) ∈ R^64
  2. 在线K-Means聚类：检测隐藏状态是否自发形成稳定的"吸引子"
  3. 转换频率矩阵：追踪状态之间的转移模式
  4. 符号稳定性指标：量化内部表征的结构化程度

理论依据：
  如果CfC在没有语言介入的情况下，仅通过感官-情绪闭环学习，
  其隐藏状态空间中自发涌现出离散的、可重复访问的区域，
  那么这些区域就是"原始符号"——无需人类语言的内部范畴。

  这是实验A vs 实验B最关键的比较点：
  - 实验B的符号空间被语言锚定（语言是外部范畴化工具）
  - 实验A的符号空间（如果存在的话）完全是内生的
"""

import torch
import numpy as np
import json, os, datetime, threading
from collections import deque

# ═══════════════════════════════════════════════════════════════
#  Means
```

Padlet Drive

symbol_tracker.py

Step 15 · Closing the perception→emotion loop: the feedback / expression subsystem

The old system only closed half the loop — *perception→emotion*. It could *feel* but never *express*, and never *grew into someone*. After Step 14's clean restart verified, I added feedback_v2.py to complete *perception→emotion→feedback→(acts on world)*, in four parts:
(1) a **staged personality** (5 attachment dimensions) slowly shaped by experience on top of innate temperament;
(2) **language acquisition** picking up words from the world's narrative as development advances;
(3) a **psyche readout** emitting one line of inner state every frame;
(4) **behavior feedback** that fires an embodied action only when metrics cross a threshold.

Key trade-off: all four are **slow variables + threshold-triggered**, not "perform every frame" — I wanted something that mostly just *lives quietly* and only occasionally says or does something.

ID

AL-M17

Status

LOAD-BEARING

0

Step 16 · Five-dimensional attachment personality: innate temperament vs. grown character

Personality isn't hard-set from birth. I separated **temperament** (the biological baseline in the GENOME — fixed) from **character** (temperament + experience — it grows), following Thomas & Chess.

Character is five 0–1 dimensions:
secure attachment / exploration courage / emotion regulation / social trust / frustration tolerance (Bowlby, Ainsworth).

Temperament sets the *starting point* (e.g. slow fright-recovery → low initial emotion regulation), then each frame nudges it by a tiny step (LR=0.0006): safety+connection high → secure attachment rises; prolonged separation → it falls; a "character snapshot" is settled at each developmental stage.

The step is deliberately tiny — character is a slow variable that only shifts under consistent long-term experience. Character also *feeds back* onto expression: below 0.4 secure attachment, it speaks more hesitantly.

ID
AL-M18

Status
LOAD-BEARING

0

Step 17 · Language acquisition: from "hearing" to "using"

Language isn't a switch, it's a **curve**. I built an 8-tier lexicon (~250+ words) following the typical order of child language development: address/need reduplications → sensory states → family & objects → food & animals → verbs → adjectives/emotions → time & place → abstract/psychological.

Two acquisition channels: **ambient exposure** (a tiny baseline gain) + **narrative grounding** (an extra boost when a word is actually "heard" in the world's narrative — grounded words are learned faster).

The more meaningful the context (high connection / strong emotion), the better a word sticks. Tiers unlock by developmental progress; on resume it *pre-seeds* early words that "should already be known" for its current age.

Expression advances through stages: prelinguistic (pure bodily tone + vocalizations) → babbling → telegraphic → simple → complex sentences (Brown 1973).

> *Note: this is a representative usable-vocabulary curve that grows with age, not a full count of the thousands of words a real 6-year-old has — early tiers already contain every word the inner monologue needs, so expression is usable at any stage.*

ID
AL-M19

Status
LOAD-BEARING

0

Step 18 · Psyche readout + embodied behavior: making the inner life legible

Two things make "what it's thinking" legible:

- **PsycheReadout**: each frame renders (emotion + body + personality + language level) into one line of inner state. Wording is gated by language stage — prelinguistic is a bodily tone like "〔waah—〕a sharp clench," and only at the complex-sentence stage does it become "There's a place inside my body that keeps hurting; I need someone to help me." Seven "feeling intents" (seek-comfort / curious-explore / wary / overwhelmed / joy-connect…) come first, then get verbalized to the current language level.
- **BehaviorSelector**: 12 embodied behaviors (crying, fist-sucking self-soothe, reaching, turning to sound, freezing…) that *don't* fire every frame — only when emotion + body budget cross a threshold, with a 6-frame refractory period to avoid repeats. Behavior is personality-modulated too: it self-soothes only with high emotion regulation, and reaches out only with high exploration courage.

ID
AL-M20

Status
LOAD-BEARING

0

Step 19 · A multi-agent living world: caregivers become subjects with internal state, not puppets

The biggest addition. NPCs used to be "an LLM playwright improvising, out of nowhere, whether the mother attends to the baby this moment" — uncontrollable and inconsistent. I rebuilt each caregiver as a **subject with internal state that evolves over time/events**:

- **Mother 陈若筠** (primary caregiver, fully modeled): fatigue / patience / anxiety / mood / friction / vulnerable moments, plus sleep debt (with a realistic ceiling where "the more exhausted, the more forced micro-sleep"), a menstrual cycle, and decaying in-law friction. Each frame, from persona + state + perceived infant behavior, she outputs: **whether she notices** the need (high awareness of loud crying, low awareness of implicit emotional needs = inferior Fe), **whether she'll respond** (anxious-avoidant: observe first), **response quality**, and **inner monologue**.
- Father (present weekday evenings), Grandmother (Saturdays only, old "pick him up the moment he cries" view that clashes with the mother's). Key design: this appraisal is injected as a **hard constraint into the playwright LLM** — "if a caregiver didn't notice / won't respond, don't let them happen to soothe the baby." State evolution is all rule-based (zero extra LLM calls). **The most important principle: never reverse-engineer growth from outcomes.** If the infant's connection can't rise because that's the necessary consequence of the mother's persona, then let it stay low. The criterion is "does the reaction fit the persona," not "is the outcome good."

> *This principle blocks post-hoc outcome tuning cheating — but it does **not** block prior bias. See reflection AL-I6.*

ID
AL-M21

Status
LOAD-BEARING

0

Closing · Final-state readout of the nurtured run at age 2 (with caveats)

Final state of the clean rerun at virtual **age 2y0m (dev_progress 100%, maturity)**:

- **Language**: complex-sentence stage, usable vocab **282**. Last inner monologue: "There's a place inside my body that keeps hurting; I need someone to help me."
- **Personality (5 dims)**: exploration courage 0.611, emotion regulation 0.545, **secure attachment 0.0, social trust 0.0, frustration tolerance 0.0**.
- **Emotion**: valence −35, arousal 73, safety 59, connection **26.9**, separation phase **3 (max)**.
- **Mother**: patience 0.07, anxiety 0.72, mood −0.97, friction 0.99; this turn she **did not notice** the need and won't respond.

I'm deliberately not treating this as a "finding" — the reason is in the next card (AL-I6**).**

Two technical notes only: (1) three dimensions hit **exactly zero** (clipped to the floor) while the two that don't depend on connection survived — that looks more like **metric degeneration / a floor artifact** than "a specific person grew." (2) Marked RUNNING, not finished: unresolved bugs remain and the zeros may be a measurement artifact, so this is not a verdict.

ID
AL-I5

Status
RUNNING

0

Reflection: The conclusion was baked in, on a leading caregiver prompt, and how to avoid it

On the previous card, "it learned language but never grew secure" — **that sentence is trivial**, because I authored the conclusion; it did not emerge. Three pieces of evidence:

(1) The mother's persona is itself a textbook *cause* of insecure attachment (anxious-avoidant, delayed response, inferior Fe / blind to emotional needs).
(2) The personality update rule then hard-codes the *effect*: secure attachment rises only when safety+connection are high and falls under separation; the mother rarely meets emotional needs → connection stays <45 → the "+" branch almost never fires; separation phase = 3 → it's debited every frame. The conclusion is *deduced* from (persona)+(rule), both written from the same theory — a tautology, not a test. (3) Smoking gun: three dimensions hit exactly zero while the two not tied to connection survived — "personality" degenerated into a deterministic function.

The half I got right: the code explicitly forbids "reverse-engineering growth from outcomes," which blocks post-hoc tuning.

But it can't block prior bias — I hand-picked a mother destined to produce insecurity, paired with a ruler that translates her behavior directly into an "insecurity" score.

How to avoid it (next version):
(1) Don't let one theory write both the world and the ruler — give caregivers only low-level mechanical params (response delay/frequency/quality) and derive attachment from interaction history via an independent measure.

(2) Treat caregivers as a *spectrum of conditions* (≥3–4 archetypes: responsive-secure / anxious-avoidant / inconsistent / over-responsive) so the result becomes a *gradient* of security vs. responsiveness, not a single foregone point.

(3) Multiple seeds per condition (N>1); report distributions.

(4) Pre-register + blind/randomize temperament before seeing outcomes.

(5) Measure what you *didn't* write into the rules — behavioral markers like proximity-seeking after separation (a Strange-Situation-style probe), not the scalar you incremented yourself.

(6) Reframe the claim: this is not evidence about "can AI attach," it's **design fiction / a consistency simulation** — "compile theory T into rules+persona and the apparatus reproduces T's predicted trajectory." The value is the apparatus and the questions.

ID
AL-I6

Status
LOAD-BEARING

0

🥚 PLAN A

Experiment A · Wild Type Architecture — Core Decisions
Experiment A is the control group — a cognitive entity that has never been exposed to language, where all internal representations self-organize purely from physical sensor data. It shares the CfC emotion core architecture with Experiment B but strips away all language and narrative layers.

Why do we need this experiment? Language is the most powerful "categorization tool" in human cognition — it slices the continuous sensory stream into discrete concepts ("cold," "hot," "mother," "danger"). Experiment B's AI is immersed in language from birth, making it impossible to determine whether its internal structure emerged spontaneously or was forcibly shaped by linguistic categories. Experiment A, by completely removing language, addresses a fundamental question: can a pure sensory-emotional loop spontaneously produce internal categories without any external symbol system? If so, will these categories resemble those delineated by human language, or be entirely alien? This directly engages the core debate around the Symbol Grounding Problem.

ID
AL-S4

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Plan A system architecture

ID

AL-S5

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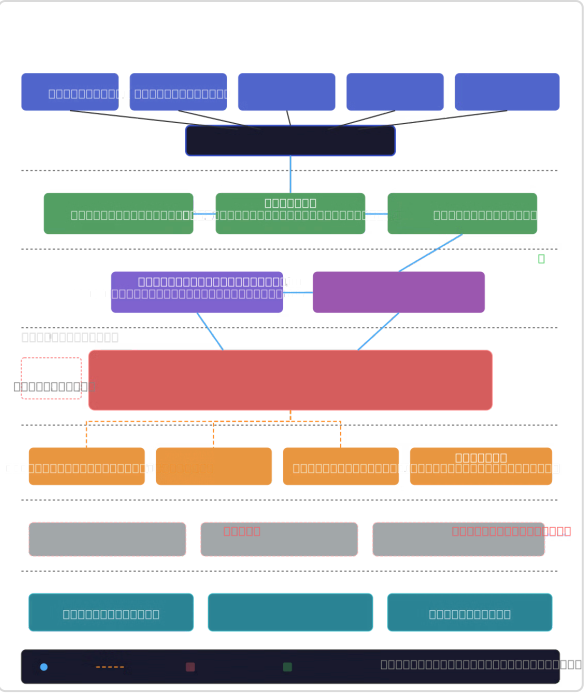

Perception Source — Arduino Physical Sensors Replace LLM-Simulated World

8-channel physical sensing (vs Experiment B's 15 channels):
illuminance (BH1750),
sound level (MAX9814),
temperature, humidity (DHT22),
human presence (HC-SR501),
pressure (FSR402),
motion (MPU6050 accelerometer),
novelty (computed from rate of change).

Why real sensors instead of simulated data? Experiment B's world is generated by an LLM playwright — essentially a "narrative reality" where every event has causal logic, character motivation, and temporal choreography. But the real physical world is full of meaningless noise, random coincidence, and pure chaos without narrative structure. Using real sensors means Experiment A faces "raw reality" rather than a simulation pre-filtered by human cognition. This distinction is critical: Experiment B's understanding of the world may inherit human biases embedded in LLM training data, while Experiment A has no such contamination source.

Why only 8 channels instead of 15? Taste and smell are extremely difficult to accurately simulate with consumer-grade sensors (electronic tongue/nose have poor accuracy and high cost). Rather than polluting the learning process with low-quality pseudo-data, we honestly remove these channels. This also creates a meaningful experimental variable: how does sensory channel deprivation itself affect cognitive development? Will a "sensory-impoverished" being develop compensatory mechanisms — such as hypersensitivity in its remaining channels?

ID

AL-S6

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Step1：CfC Neural Core — 22D Input vs 36D

Input dimensions:
sensory(8) + prediction_errors(8) + body_budget(3)
+ autonomic(1) + circadian(2) = 22D.
The CfC network structure is identical to
Experiment B (64 AutoNCP neurons, 5D
emotion output), with fewer input ports.

Why keep the same network architecture? This is the core principle of experimental design — controlling variables. If we changed both the "world" and the "brain" simultaneously, we couldn't determine whether behavioral differences stem from which factor. Both experiments use the same CfC neuron count, the same AutoNCP wiring strategy, the same 5D emotion output space — differing only in the number and source of perceptual input channels. This way, any behavioral difference can be attributed to "different ecological niches" rather than "different hardware."

```
#!/usr/bin/env python3
"""
life_wild.py — AI生命体 · 实验A 野生型 (Wild Type)
=====================================================
纯物理传感器驱动 · 无语言模型 · 无叙事层

与实验B (life_v2.py) 的关键差异:
  - 感知来源: Arduino物理传感器 (8通道) 替代 LLM模拟世界 (15通道)
  - CfC输入: 22维 (vs 36维)
  - 移除: LLMPlaywright, Worldbook, EventTranslator
  - 新增: ArduinoSensorBridge, InternalSymbolTracker
  - 保留: CfC情绪核心, 预测引擎, 身体预算, 自主神经, 昼夜节律, 基因组, Phase1-4发育

依赖:  pip install torch ncps
       pip install pyserial  (仅Arduino真实模式)
运行:  python life_wild.py              # 模拟模式
       python life_wild.py --real COM3   # Arduino真实模式
"""

import time, math, random, datetime, os, signal, sys, json, argparse
import torch
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim

# Windows GBK 终端 UTF-8 输出
if hasattr(sys.stdout, 'reconfigure'):
    sys.stdout.reconfigure(encoding='utf-8', errors='replace')
r, 'reconfigure'):
igure(encoding='utf-8', errors='replace')
```

Padlet Drive

life_wild.py

ID
AL-M22

Status
LOAD-BEARING

0

Step2：emoved Modules — Stripping All Language & Narrative

Source files (removed from Experiment B):

```
world_v2.py → class LLMPlaywright,
```

```
class EventTranslator; worldbook.py → WORLDBOOK, EVENT_TYPES
```

Three modules completely removed:
LLM Playwright thread (LLMPlaywright),
World Setting Book (Worldbook),
Event Translator (EventTranslator).

Why "complete deletion" rather than "disabling language output"? Even without letting the LLM generate language output, as long as world events are still orchestrated by an LLM, information has already been pre-organized by linguistic structure. The LLM-generated sequence "mother enters → breastfeeding → leaves" is essentially a token sequence with causal narrative — even when translated into pure numerical channels, the data still implicitly carries the causal framework of human language. Complete deletion means: the sensory data Experiment A receives contains zero information pre-structured by language. A rise in illumination might be caused by someone turning on a light or clouds dispersing — to Experiment A, this is simply "illuminance value increased," with no conceptual presupposition of "someone," "light switch," or "clouds."

What does this create? A genuinely "pre-linguistic" cognitive entity. If discrete, repeatedly accessible regions spontaneously emerge in its CfC hidden states, these would be "proto-symbols" independent of human language — an extraordinarily rare experimental condition in cognitive science.

this is an ablation — removing a component entirely in order to see what the system does without it. The reason it had to be deletion rather than disabling is stated above: a disabled LLM still leaves world events organised by linguistic causal structure, so the variable would not actually have been removed. An ablation that leaves the thing in place is not an ablation.

ID
AL-M23

Status

LOAD-BEARING

0

Step3：New Modules — Sensor Bridge + Internal Symbol Tracker

Three new modules:
ArduinoSensorBridge,
PhysicalAmbientEngine,
InternalSymbolTracker.

Why ArduinoSensorBridge? Real sensors drop packets, have latency, and drift — this isn't a bug, it's a feature. Experiment B's simulated world delivers data with 100% fidelity, but real-world perception is always incomplete. The bridge layer handles these real-world issues (packet loss retry, last-frame fill, drift calibration) while preserving the data's "imperfection." It also supports real/simulated dual-mode switching for debugging without Arduino hardware.

Why is InternalSymbolTracker the most important new module? This is the observation instrument for the entire Experiment A. It does three things: ① Online K-Means clustering — continuously monitors whether the CfC's 64-dimensional hidden states spontaneously converge into a few stable "attractor" regions (if they do, these are candidates for "proto-symbols"). ② Transition matrix tracking — if state transitions follow patterns rather than random jumping, the system has formed an internal "grammar" — a sequence structure independent of human language. ③ Symbol stability scoring — a composite 0-1 score combining compactness (within-cluster variance), separation (between-cluster distance), and temporal stability (frame-to-frame consistency), quantifying the degree of internal symbolization.

Why does this score matter? It's the key metric for answering the core research question. If symbol_score rises steadily over developmental time, the system is spontaneously developing discrete internal categories without linguistic intervention. If it remains near zero, pure sensory experience without language anchoring is insufficient to produce symbolic representation — which itself would be an important finding.

ID
AL-M24

Status
LOAD-BEARING

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Step4: Preserved Core Mechanisms — Comparable Baseline with Experiment B

All neuroscience mechanisms retained:
sensory preprocessing (habituation/sensitization + developmental gating),
prediction engine,
body budget system,
autonomic nervous controller,
circadian oscillator,
GENOME,
Phase 1-4 developmental timeline.

Why preserve each mechanism?
Sensory preprocessing (Groves & Thompson 1970 habituation/sensitization): Attenuation of repeated stimuli and amplification of high-arousal stimuli are fundamental properties of all nervous systems, independent of language. The wild type needs this mechanism to extract signal from noise.

Prediction engine (prediction error drives emotion): Predictive coding is a core operating principle of the brain (Friston's Free Energy Principle), independent of language — even animals without language continuously predict the next frame of sensory input. Prediction error is the driving force for exploration and learning.

Body budget system (Barrett 2017 constructionist emotion theory): Emotions are not "reactions" but byproducts of body budget management. The wild type also needs to manage energy/hydration/thermoregulation, but since there are no "feeding" events, energy recovers through circadian metabolic rhythms — simulating an organism that self-maintains metabolism rather than an infant dependent on caregivers.

Autonomic nervous controller (Porges polyvagal theory): The three-zone hierarchy (freeze/fight-flight/safe-social) is an evolutionarily conserved system present in all vertebrates. The wild type uses PIR infrared signals instead of narrative "social signals" to trigger ventral vagal activation — "something is nearby" rather than "mother is holding me."

GENOME (INxJ temperament parameters): Both experiments use identical innate parameters — same sensory thresholds, arousal baselines, emotional inertia, exploration bias. This ensures any behavioral differences originate from postnatal environment (with language vs. without language) rather than innate temperament. This is the logic of "identical twins raised apart" in experimental design.

ID
AL-M25

Status
LOAD-BEARING

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Hardware Decision Log

Initial instinct: The most obvious output direction for this project was an emotional-companionship device — something in electronic-pet / Tamagotchi territory. It felt like the safest fit for the CfC emotional core already built in Investigate.

Why it was rejected: Two problems. First, it overlapped thematically with the Unit 2 ADHD therapeutic toy project — same "emotional wellbeing object" territory, weak interdisciplinary distinction. Second, and more importantly, it didn't answer the actual question the field interviews had been asking — which was never "how does a person come to feel affection for a thing," but "how does a person understand what a living system is experiencing without a data dashboard."

ID
AL-I19

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Step 5: a physical body

this is a material selection record. The materials in question are sensors — the components that determine what this thing can physically feel — and the selection criterion was explicitly cost and ease of use for a beginner, not signal quality. I am naming it as a material decision because that is what it is: what the creature can feel was decided by which components I could afford and get working, not by which would have been best.

How the Plan B Sensor Set Was Actually Chosen — and What Kind of Decision This Was
this documents is a single, quick AI-assisted planning exchange, not a comparative sensor-evaluation process. The selection criterion was never "which sensor gives the richest signal" — it was cost and ease of use for a beginner, because the starting brief was explicitly "use the simplest, cheapest components, assume no prior hardware experience."

The actual logic used, in order:

1. Four target emotional dimensions were set first — arousal, safety, connection, comfort — as the internal state the system needed to track.
2. Sensors were then chosen backward from those four dimensions: for each dimension, what is the cheapest, simplest possible signal source a beginner could wire up without a supporting processing pipeline?
3. This is a reverse-engineered mapping (dimension → cheapest matching signal), not

Resulting sensor set:

Sensor	Mapped dimension	Selection basis
PIR (passive infrared) presence sensor	Connection	Cheapest, simplest binary "is someone nearby" signal available
DHT11 temperature/humidity	Comfort	Cheapest, most common, best-documented option in this sensor category
Basic sound-level sensor (noise intensity only, not speech)	Arousal	Simplest way to detect environmental stimulation without adding a speech-processing pipeline
Basic light sensor	Not clearly mapped to a single dimension	Included as a general day/night background signal

a forward evaluation (survey available sensors → judge their merits → pick the best).

Architecture this fed into: Arduino reads all sensors → sends over USB serial to a local machine → a Python script logs a periodic "perception snapshot" (e.g. light level, sound level, temperature, humidity, presence, timestamp).

Honest self-assessment: three of the four sensors (PIR, DHT11, sound) have a clear, stated one-to-one link to an emotional dimension. The light sensor does not — it was included as a general-purpose background variable without the same explicit reasoning given to the other three. This inconsistency is part of the actual record, not a gap introduced later.

ID

AL-M26

Status

LOAD-BEARING

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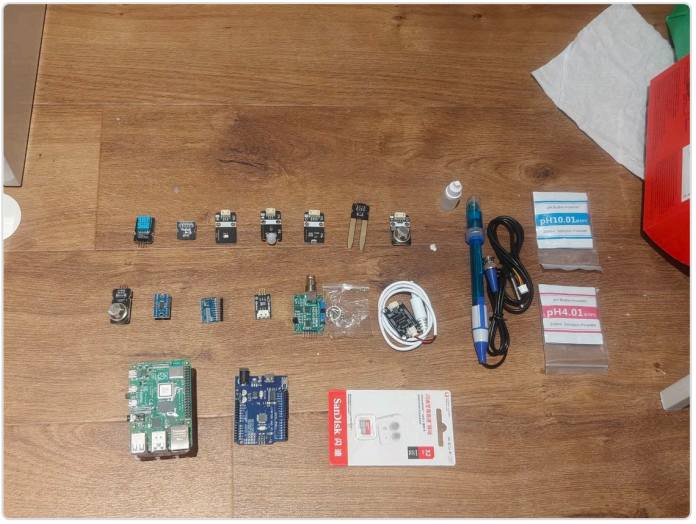
0



What Was Not Chosen, and Why

This is the material comparison for the project. Four components, each with a stronger alternative that existed and was set aside, with the reasoning recorded at the moment of the decision.

Framing note: the comparison documented here is a comparison of reasoning at the moment of proposal — which stronger alternative existed for each sensor, and why it was set aside — not a record of physically testing multiple components and rejecting some after trial. That distinction matters and is kept explicit here rather than blurred.

- **PIR presence sensor — chosen over a camera + vision-based human detection**
  A camera could have provided far richer information (who is present, what they are doing, not just whether anyone is present at all). It was not chosen because vision-based detection requires an entire additional processing pipeline — calibration, a detection model, meaningfully more compute — which was judged to be over-engineering for a beginner's starting point. PIR was chosen purely because it is a low-cost, three-wire module that outputs a simple boolean, meeting the minimum requirement ("is someone nearby") with the least possible complexity.

- **DHT11 — chosen over DHT22 or BME280**

Both alternatives offer higher measurement precision (DHT22), or additional measured variables such as air pressure (BME280). Neither was chosen, because at this starting stage precision was not the problem being solved — reliably getting a reading at all, with the most common tutorials and the most mature Arduino library support, was the actual priority. This was a pure convenience/documentation-availability choice, not a technical merit judgement.

- **Sound-level sensor — chosen over a microphone with speech recognition**
  A microphone with speech recognition would have enabled genuine language interaction, which is a meaningfully more capable option. It was explicitly proposed as an optional addition rather than part of the core set, on the stated reasoning that adding speech recognition shifts the project's scope from "environmental perception" into "language processing" — a deliberate scope boundary, not an oversight.

- **Light sensor — no comparable alternative was proposed at all**
  Unlike the other three, no stronger alternative was named or reasoned through for the light sensor at the time. It was added as a general day/night background signal without being tied to a specific emotional dimension the way the other three sensors were. This is flagged here as a weaker point in the original reasoning, not retrospectively strengthened.

the strength of this initial plan is that it prioritised getting a working perception loop running quickly, in a form usable by someone with no hardware background — not that it represents a rigorously benchmarked sensor selection. Documenting it this way is more accurate.

ID
AL-I20

0

Step 6: Write the program the sensor board itself runs
File: `sensor_firmware.ino`
What was done: Wrote the small program that lives on the sensor board itself. Its entire job is: read every sensor in turn, pack the readings into one line of plain text, send that line down the cable, wait 2.5 seconds, repeat. Written first for the larger of the two boards I had available (Arduino Mega 2560).

Why: Before anything can be built, the physical side and the software side need an agreed contract — what gets sent, in what format, how often. Fixing that first means the two halves can then be developed separately and by different logic.

What it means: Two decisions are hidden in this that are not technical.

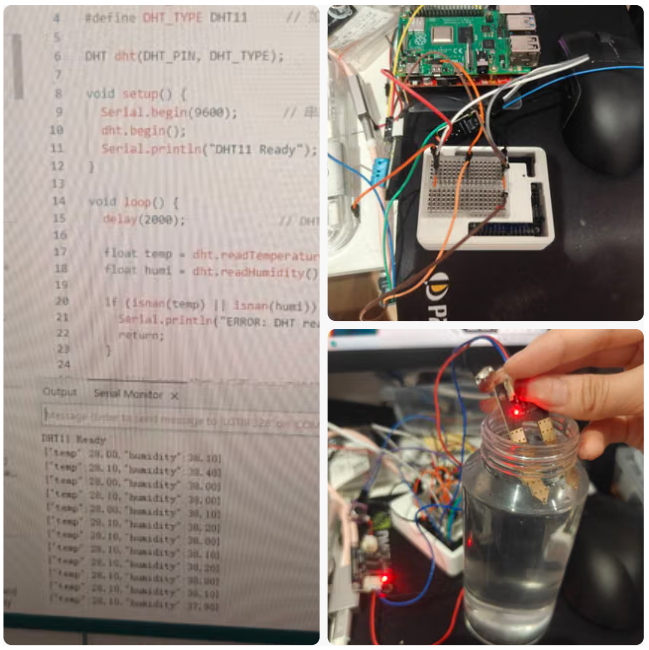
- First, the board samples every 2.5 seconds while the software runs on a 3-second cycle — deliberately slightly faster, so the software never has to wait on a reading that hasn't arrived.
- Second, and more interesting: **that 3-second cycle is the length of the creature's present moment.** Everything that happens inside one cycle is, to it, simultaneous. I chose that number. Nothing in biology told me to.

ID
AL-M27

Status
SUPPORTING

0

0



Step 7: Port the same program onto the cheap beginner board
File: `sensor_firmware_uno.ino`

What was done: Rewrote the same program for the Arduino UNO — the smallest, cheapest, most standard beginner board. The text it sends out is byte-for-byte identical to the previous version, so the software on the computer needs no change at all.

Why: Consistency with the stated brief. The whole Plan B sensor list was chosen on the principle "cheapest and simplest thing a beginner can actually get working" (see `AL-M26`). Specifying an expensive board for that same set would have contradicted the reasoning I'd just written down.

What it means: The practical discovery was that two of the six sensors — the light meter and the movement sensor — can share the same pair of wires, because each answers to a different address on that shared line. That is the only reason six sensors fit on a board this small.
The more useful lesson is about the shape of the decision: **I kept the output format frozen so the hardware could change without the experiment changing.** That seam is what makes it possible to swap the entire physical layer later — including for a fake one, which is exactly what happens in Step 9.

ID
AL-M28

Status
SUPPORTING

0

Step 8: Audit the parts list — the code no longer matches the plan
What was done: Put the sensor list from the plan (AL-M26) side by side with the sensor list the code was actually written against, and found they had drifted apart. The plan named four things: presence, temperature-and-humidity, sound level, light. The code addresses six, adding **touch pressure** and **being moved** — and quietly upgraded the other four from the cheapest versions to better-specified relatives.

Why it happened: Writing the code forced a question the shopping list never asked: *what can this thing actually feel*? Once framed that way, a creature that is supposed to have a body but cannot feel being touched or being picked up looked like an obvious hole, and I filled it while writing rather than going back to re-do the selection.

What it means: This is scope drift, and I only caught it while writing this documentation up. Two whole channels entered the system **without passing through the reasoning I had applied to the first four** — no cost comparison, no alternative considered, no justification written at the time. Because nothing has been bought or wired yet, nothing is broken; the cost so far is only that my plan and my code describe different creatures. I'm recording it rather than tidying

it away, because the tidy version would make the selection process look more deliberate than it was. **The honest account is that the parts list was written by shopping logic and the code was written by empathy, and nobody reconciled them.**

ID
AL-I10

Status
LOAD-BEARING

0

Step 9: Build a fake world, so the software can be tested with no hardware at all
File: `sensor_bridge.py`

What was done: Wrote a simulator that manufactures exactly the same eight numbers a real sensor board would send, from a virtual clock: light following a day/night curve (sunrise 05:50, sunset 18:20, near-black at night, brightest early afternoon), background noise quiet at night and louder by day with a 3% chance each moment of a sudden burst, indoor temperature drifting slowly around 25.5°C, humidity cycling, **someone walking past every 5–40 minutes and staying between 30 seconds and 10 minutes**, an occasional press on the pressure pad, and near-total stillness broken by the odd nudge when someone is there.

Why: The board wasn't built. Waiting for hardware before testing any of the software would have cost weeks of experiment time for no gain, and it would have left me unable to tell a hardware fault from a software fault when things eventually went wrong.

What it means: The simulator and the real board are deliberately interchangeable — both answer the same single instruction and return the same eight numbers, so the rest of the system cannot tell which one it is talking to. That is a genuine engineering virtue and also the thing I have to be loudest about, because it means **everything reported from here on was run against this invented world, not a real room.** I made the system stamp `"mode": "sim"` into every single line it logs, so no future reader — including me — can mistake one for the other.

simulate first, and make provenance non-optional. Every line this system logs is stamped "mode": "sim", so no future reader — including me — can mistake invented data for a real room. The method is not "simulate"; it is *simulate, and make it impossible to forget that you did.*

ID
AL-M29

Status
LOAD-BEARING

0

Step 10: Deliberately make the fake world boring — and notice the trap in that

What was done: Made a rule that the simulated world contains **no events, no people with intentions, no cause and effect, and no story of any kind.** Nothing in it means anything. Light goes up and down, noise happens, a warm body occasionally passes through. Nobody arrives *in order to* do anything.

Why: This is the entire reason the wild type exists. The nurtured experiment's world was written by me in language — a mother, a father, a grandmother, events with intent and consequence. If the wild type's world contained even a trace of the same authored meaning, it would stop being a control group and become a second version of the same experiment.

What it means: It does the job it was built for, and it contains a trap I want on the record: **a world I wrote to be meaningless is still a world I wrote.** I chose how long the day is, how often a visitor comes, how loud the room is, how much things fluctuate. So if nothing structured ever emerges in the creature, one completely honest explanation is that my invented world simply had no structure in it to find. From inside a simulation I cannot separate "the creature failed to find structure" from "there was no structure to find." That single sentence is the strongest argument in this whole project for actually building the

physical board — not because real sensors are more impressive, but because **a real room is the only world I can't be accused of having authored.**

ID
AL-I11

Status
LOAD-BEARING

0

Step 11: Add an eighth channel that is not a sense at all
File: `sensor_bridge.py`

What was done: Added a "novelty" channel alongside the seven real ones. It isn't measured by anything. It is calculated: how much did all the other seven readings change since the last moment, averaged.

Why: I wanted the creature to be able to react to *change*, not only to levels. A room that is steadily bright and a room that has just become bright are, in terms of what a nervous system should do next, completely different situations — but they can look identical if you only read the current brightness.

What it means: This is the one place in an experiment premised on *not* pre-loading meaning where **I pre-loaded meaning.** I decided, on the creature's behalf and before it ever ran, that "how much things just changed" is a quantity worth feeling. There is decent justification for that — startle and orienting responses in real animals work on exactly this — but it is still my judgement inserted into the perception layer, not something the creature derived. I'm keeping it, and naming it, rather than letting it sit in the code disguised as a sensor.

ID
AL-I12

Status
SUPPORTING

0

Step 12: Decide how raw physics turns into feeling
File: `life_wild.py`

What was done: Wrote the rules that convert the eight incoming numbers into the five internal feelings the system tracks. In plain terms: **connection is driven purely by whether a warm body is detected nearby** — present and it climbs, absent and it decays over roughly the next two and a half minutes down to a floor. **Safety** is damaged by sudden loud noise, by being moved abruptly, by darkness, and by temperature outside a liveable band. **Arousal** rises with how much everything just changed, weighted so that sound and someone's presence matter most. **Comfort** is best near 25°C and with gentle pressure — too much or too little of either makes it worse.

Why: There is no caregiver in this experiment and no language, so nothing can ever tell this creature how to feel. Its entire emotional life therefore has to be derivable from physics alone. Writing these rules is what makes that possible.

What it means: I wrote these rules, and this is the point where the wild type is least wild. The honest framing is not that I removed my authorship — it's that **I moved it.** In the nurtured experiment I authored the creature's stories; here I authored its reflexes. What is genuinely different, and worth defending, is the *kind* of authorship: these rules are general and stimulus-driven, stated once in advance and applied identically to every moment, rather than me narrating events one at a time and deciding what each one should mean. That is a weaker form of authorship, but it is not none, and claiming otherwise would be the most tempting lie available in this project.

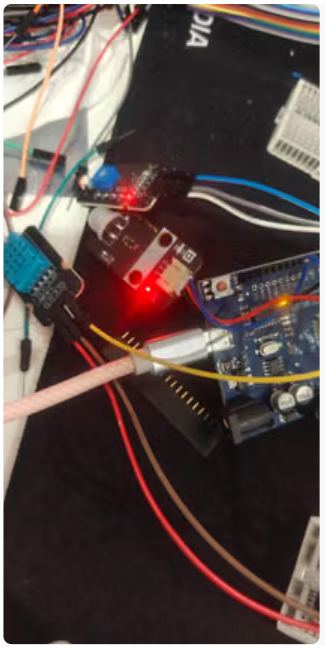
ID
AL-M30

Status

LOAD-BEARING

0

0



Step 14: Build the instrument that will decide whether the experiment worked
File: `symbol_tracker.py`

What was done: Built a separate observation tool that watches the network's internal state from the inside. Every so often it takes a snapshot of that state, sorts the snapshots into groups of similar ones as they arrive, and then tracks two things: which group the creature is currently in, and which group tends to follow which. It scores the result on how tight the groups are, how well separated they are from each other, and how stable the creature is within one group over time. It also writes down the most common sequences it sees, in the form `state 6 → state 6 → state 6`.

Why: The research question is whether internal categories can form without language. The creature has no language, so it can never tell me what it is doing — which means the only way to answer the question is to read its internal state directly and look for grouping. Without this tool there is no experiment, only a program that runs.

What it means: Two things. First, this is the actual measuring instrument of the entire project — the graphs and the creature are secondary to it. Second, and more importantly as a matter of method: **I built it before the run rather than after.** If I had waited until I'd seen the data, I would have been free to define "success" as whatever pattern happened to show up. Committing to the measure in advance is what makes a negative result mean something — and, as Step 21 shows, it also gives me something specific enough to later catch as wrong.

this is pre-registration in miniature — the measure is committed to before the data exists. Its value is not rigour for its own sake. It is that a negative result becomes capable of meaning something, because I cannot retrofit "success" onto whatever

ID
AL-M32

Status
LOAD-BEARING

0

Step 15: First run — and it crashed in the middle of the night
File: `growth_milestones_wild.jsonl`

What was done: Started the first full run at **23:51:38 on 4 July**, at 25× speed so that days of virtual development pass in hours of real time. At **02:26:39** the process died and I restarted it, resuming from where it had got to — one virtual year of development, 77,000 steps in.

Why record a crash at all: Because a resumed run is not identical to an uninterrupted one, and anyone reading the results is entitled to know there is a seam in them.

What it means: The restart is written into the log as a second start event, so the seam is visible rather than papered over. The wider point is one I keep having to relearn: **the log has to record the accidents, not just the intended events.** If I had restarted quietly, the file would have described a clean continuous run that never happened, and every later conclusion would have rested on a small lie I'd have forgotten telling.

ID
AL-M33

Status
SUPPORTING

0

Step 16: Discover the run had quietly overshot — nothing told it to stop
File: `archive_pre_rerun_20260705_144331/growth_status_wild.json`

What was done: Let the run continue and only afterwards noticed it had reached a virtual age of **6 years** by 14:32 on 5 July — over 433,000 steps. The nurtured experiment had been stopped at **2 years**.

Why it happened: There was no stop condition in the code. I had never written one, because I had never explicitly decided how long the experiment should be. It simply kept going because nothing told it not to.

What it means: The two experiments were no longer comparable, and the whole purpose of the wild type is to be comparable — it exists to isolate what difference language and caregiving make, which only works if everything else, including duration, is held equal. I found this by accident while reading the numbers, not by design. **The omission is more interesting than the fix: I had specified in enormous detail what the creature would be, and not at all when the experiment would be over.** That is a very recognisable failure mode — obsessing over the object and forgetting to define the study.

ID
AL-I13

Status
DEAD END

0

Step 17: Add a stopping rule and throw away a finished run
File: `growth_milestones_wild.jsonl`

What was done: Wrote an explicit end condition — stop when development is complete, at roughly 2 virtual years — archived the entire 6-year run rather than deleting it, and re-ran **both** experiments from zero. The clean run started **14:43:56** and finished **19:32:31** on 5 July, ending exactly at 2 years and 144,000 steps, with the reason recorded in the log as "reached developmental cap."

Why: A comparison between two things that ran for different lengths of time isn't a weak comparison, it's not a comparison at all. Any difference found could be explained by the extra four years rather than by anything I was actually testing.

What it means: This cost several hours of real compute and meant discarding a completed run, which was genuinely hard to do — the finished thing sitting there is very persuasive when you're tired. I archived rather than deleted it, which turned out to matter enormously: **that discarded run becomes the single most informative piece of evidence in the whole experiment** (Step 20). The decision that felt like admitting failure is the one that produced the finding.

ID
AL-M34

Status
LOAD-BEARING

0

Step 18: Read the result — and notice three of the five feelings had stopped moving
File: `wild_log.jsonl`

What was done: Went through the last 5,000 recorded moments of the clean run one by one instead of looking at the summary. **Arousal read exactly 73.1 in every single one of them. Connection read exactly 26.9 in every single one. Seeking read exactly 26.9 as well.** Only two of the five feelings — comfort and safety — were still varying at all.

Why this is not a mood: Two details give it away. First, 73.1 and 26.9 sit perfectly symmetrically either side of the midpoint of the scale, 50 — they are 23.1 above and 23.1 below. That is the fingerprint of an output that has been pressed flat against its own ceiling and floor, not of a temperament. Second, connection had already been stuck at 26.9 at the **very first developmental milestone**, at a virtual age of four and a half months, and never moved again for the remaining year and a half.

What it means: I had a beautiful sentence ready — "*the wild type matured into a state of permanent vigilance and disconnection.*" It reads well, it fits the theme of the project, and it is false. What actually happened is that its emotional output saturated and stopped carrying information. **The most dangerous moment in this whole experiment was the one where the wrong answer was also the more poetic one.** The only thing that saved it was reading individual rows rather than the summary.

ID
AL-I14

Status
LOAD-BEARING

0

Step 19: The decisive check — it had stopped responding to the sensor meant to drive it
File: `wild_log.jsonl`

What was done: Split those same 5,000 moments into the ones where a body was detected nearby and the ones where it wasn't, and compared connection across the two groups. Presence was detected in **1,628 of them — a third of the time**, so there was plenty to compare. Result: **connection averaged 26.9 when someone was there, and 26.9 when nobody was.** Identical to the last decimal place, with no spread at all in either group.

Why this is the test that matters: Connection was defined, in Step 12, as being driven *purely* by presence — that is its only input. So this comparison directly asks whether the wiring between the physical world and the creature's feelings is still alive. If the input changes a third of the time and the output never moves, the link is dead, and no amount of looking at a graph would have established that.

What it means: This is the most useful five minutes of analysis in the project, and I only ran it because a suspiciously round constant made me distrust my own result. **A number that never changes should be treated as a broken instrument until proven otherwise, not as a finding.** The temptation runs entirely the other way: a flat line looks like stability, stability looks like a personality, and a personality is a much more attractive thing to write about than a bug.

ID
AL-I15

Status
LOAD-BEARING

0

Step 20: Compare against the discarded run — the cap had destroyed the phenomenon
File: `archive_pre_rerun_20260705_144331/wild_log.jsonl`

What was done: Ran exactly the same presence-versus-connection test on the 6-year run I had archived in Step 17. There, the link was clearly **alive**: connection averaged **54.8** when a body was detected and **27.3** when none was — a large, consistent difference, with connection ranging up to 56.3. Then checked that same run at the 2-year mark: connection was still pinned at 26.9, exactly as in the capped run.

Why this changes the conclusion: It means the coupling between the physical world and the creature's feelings was never broken by design. It simply **took longer than two virtual years to appear** — and the stopping rule I added in Step 17, for entirely good reasons, ended the experiment before the only interesting thing in it had started to happen.

What it means: This is the single most useful thing I learned in the whole project, and it is uncomfortable. A methodological decision made to protect the comparison — hold duration equal — was itself a bias, because **it assumed the two creatures develop on the same clock.** A child given language and constant response should be expected to develop faster than one left alone in a room with no language at all; forcing them onto the same timetable guarantees the slower one looks empty. Fairness in a comparison is not the same as identical treatment.

Honest limit: the log file only keeps the last few thousand moments and overwrites the rest, so I have roughly the final ten minutes of each run and nothing in between. I can state that the coupling was absent at 2 years and present at 6. **I cannot say when it appeared, and that is my instrumentation failure, not a property of the creature.**

ID
AL-I16

Status
LOAD-BEARING

0

Step 21: Audit the success score — and find it rewards the system for dying
File: `symbol_log.jsonl`

What was done: Opened up the headline number. The run finished with a symbol score of **0.64** and a structure rating of **0.94**, which reads like a success. Underneath: **one single active group**; tightness scored 0; separation scored 0.0; **stability scored the maximum, 1.0**; the most common sequence recorded was `state 6 → state 6 → state 6`, and 44,029 of the sorted moments sat in that one group. The creature never left state 6.

Why the score went up as the system went flat: Stability was scoring maximum **because** nothing ever changed. The structure rating was high **because** a system with one state is perfectly predictable. Both components were measuring exactly what I wrote them to measure, and what I wrote them to measure turned out to be satisfiable by stopping.

Two further faults found in the same file: the tracker's own record gives the score as **0.30** at the same moment the run's headline reported **0.64** — two different numbers for one quantity, differing by a factor of two, and I hadn't noticed. And the tracker only wrote **two entries** in the entire 144,000-step run, the last at step 47,500: **the measuring instrument stopped recording a third of the way through and nothing alerted me.**
What it means: A metric that awards its highest marks to a system that has stopped is not measuring what I claimed it measured. I built it to detect stable attractors, and it faithfully detected the most stable attractor available: death. The fix is not a better formula — it is a requirement I should have imposed at the start: **any claim of "structure" has to survive a stated null case, and a frozen system must be constructed to score badly.** Until a measure can fail, a good result from it means nothing.

ID
AL-I17

Status
LOAD-BEARING

0

Step 22: What the wild type actually shows, and what it does not

What the result is, stated exactly: Run against a simulated environment I wrote myself, and stopped at two virtual years, the wild type **did not develop distinguishable internal states.** It collapsed into a single internal state, and three of its five emotional outputs saturated and stopped carrying information. A longer run of identical code, without the stopping rule, showed a clear live coupling between physical presence and the connection dimension by six virtual years — but even that run only ever reached two internal groupings, not a repertoire, and I cannot say when the coupling appeared.

What this does not license me to say: It does **not** show that internal categories can't form without language. There are at least four explanations I currently have no way of separating: my invented world may genuinely contain no structure to find; two years may simply be too short; the emotional output saturates and may be destroying the signal before the tracker can see it; and the tracker's score, as Step 21 shows, cannot yet tell structure apart from stasis. Any one of those would produce exactly the result I got.

What I would do next, in order: actually build the six-sensor board and run it in a real room, so the world is one I can't be accused of authoring; drop the shared stopping rule and let each experiment run to its own natural end, reporting the different durations rather than forcing them equal; fix the output saturation so the feelings can still move; and redefine the symbol score so that a frozen system scores zero by construction.

What it means: The wild type's job in this project was to be the control that shows how much of the nurtured type's development came from being spoken to and responded to. **It cannot yet do that job.** Saying so plainly is more useful than any graph I could produce from this data, because the whole comparison the project rests on is currently unearned — and knowing exactly which four things stand between me and earning it is a better position than a chart that implies I already have.

ID
AL-I18

Status
LOAD-BEARING

0

FUKT SOUND

Bought on an RCA bookshop visit. The entire issue is about drawing sound — taking something with no inherent visual form and giving it one. That is precisely the problem the device's expression layer has to solve: an emotional state has no natural appearance, so whatever visual form it takes is an authored translation, not a readout. Kept here as a reminder that the LED mapping is a design decision I am accountable for, not a neutral display of data.

ID
AL-C3

Status
SUPPORTING

0

Joydeb Roaja, Go Back to Roots (2020)
— organs drawn as root systems

Tate, seen the same day as the Long. Roaja renders human anatomy as entangled root, vine and network structures — a body drawn as a system of connections rather than as flesh. Since this device's sensing "organs"

are literally probes extending into soil and water, that substitution is not metaphorical here; it describes the actual thing being built.
This prompted an alternative direction for the expression layer: making the entity's organs in wool felt and yarn — soft, hand-made, biological — rather than housing them in a smooth printed shell. Combined with the FUKT issue on drawing sound, the thought went further: felt driven by servos or similar actuators, so the diagram physically moves. Not a light changing colour to represent a state, but a soft form that contracts, tenses and expands — a kinetic diagram rather than a graphic one.

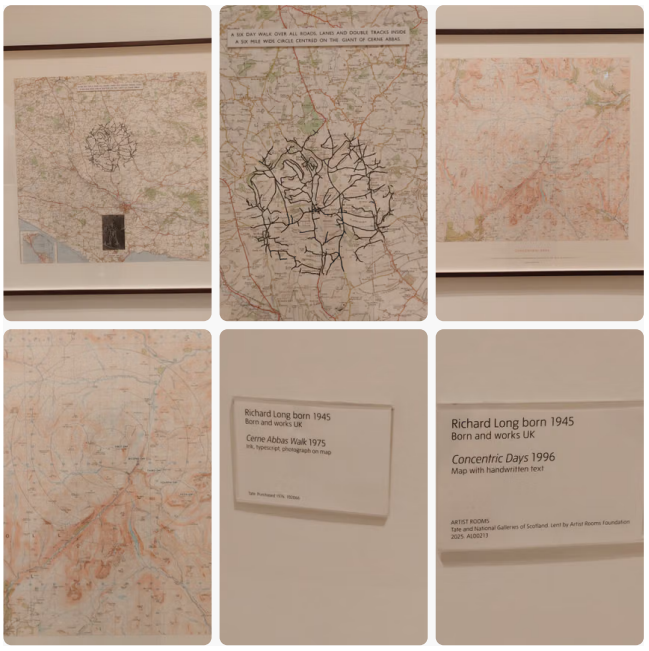
Material note: What this proposes is a material substitution: wool felt and yarn — soft, hand-made, biological — in place of a printed shell, driven by servos so the form contracts and expands. A kinetic diagram rather than a graphic one.
Status note. At the end of Investigate this was a live possible outcome, not a discarded one — which is what Investigate is meant to produce. It was closed later, by a decision made in Realise, and the reason is recorded there rather than here. See RE-06.

ID
AL-C5

Status
SEED

0

Richard Long, Cerne Abbas Walk (1975) — a rule fixed in advance, and the work is whatever running it produces
Tate, seen while both experiments were running. What I took was the method, not the image.



Long fixes an arbitrary constraint before he starts — a set radius, a set number of days — walks it, and the work is the trace that execution leaves behind. He does not choose the result. He chooses the rule, and is then bound by it.

Where this lands in my own work: AL-M32. I built the symbol tracker before the run rather than after, and the reason I gave there was that if I had waited until I had seen the data, I would have been free to define "success" as whatever pattern happened to show up. That is structurally Long's radius. Committing to the measure in advance is what makes a negative result mean anything — and, as AL-I17 shows, it is also the only reason I was later able to catch the measure itself as broken.

I am not claiming Long as an influence on the experiment; the tracker was already written when I saw this. What the work gave me was a name for something I had done for narrowly methodological reasons and had not recognised as a form. A rule executed becomes the artefact.

Separately, and more practically, it also suggested how to present Plan A and Plan B: not as screenshots of software running, but as a single static artefact holding an entire lived duration at once, legible without needing to be watched in real time.

ID

AL-C4

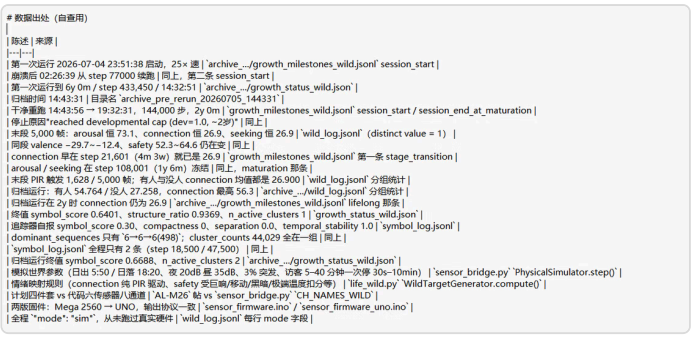
Status

SUPPORTING

0

DEVELOP LOG & DATA SOURCES

0



数据出处（自查用）

路述	来源
第一次运行 2026-07-04 23:51:38 启动，25× 速	`archive_.../growth_milestones_wild.jsonl` session_start
崩溃后 02:26:39 从 step 77000 续跑	同上，第二条 session_start
第一次运行到 6y 0m / step 433,450 / 14:32:51	`archive_.../growth_status_wild.json`
归档时间 14:43:31	目录名 `archive_pre_rerun_20260705_144331`
干净重跑 14:43:56 → 19:32:31，144,000 步，2y 0m	`growth_milestones_wild.jsonl` session_start / session_end_at_maturation
停止原因"reached developmental cap (dev=1.0, ~2岁)"	同上
末段 5,000 帧：arousal 恒 73.1、connection 恒 26.9、seeking 恒 26.9	`wild_log.jsonl`（distinct value = 1）
同段 valence −29.7～−12.4、safety 52.3~64.6 仍在变	同上
connection 早在 step 21,601（4m 3w）就已是 26.9	`growth_milestones_wild.jsonl` 第一条 stage_transition
arousal / seeking 在 step 108,001（1y 6m）冻结	同上，maturation 那条
末段 PIR 触发 1,628 / 5,000 帧；苍人与没人 connection 均值都是 26.900	`wild_log.jsonl` 分组统计
归档运行：有人 54.764 / 没人 27.258，connection 最高 56.3	`archive_.../wild_log.jsonl` 分组统计
归档运行在 2y 时 connection 仍为 26.9	`archive_.../growth_milestones_wild.jsonl` lifelong 那条
终值 symbol_score 0.6401、structure_ratio 0.9369、n_active_clusters 1	`growth_status_wild.json`
追踪器自报 symbol_score 0.30、compactness 0、separation 0.0、temporal_stability 1.0	`symbol_log.jsonl`
dominant_sequences 只有「6→6→6(498)」，cluster_counts 44,029 全在一组	同上
`symbol_log.jsonl` 全程只有 2 条（step 18,500 / 47,500）	同上
归档运行终值 symbol_score 0.6688、n_active_clusters 2	`archive_.../growth_status_wild.json`
模拟世界参数（日出 5:50 / 日落 18:20、夜 20dB 量 35dB、3% 突发、访客 5–40 分钟一次停 30s–10min）	`sensor_bridge.py` `PhysicalSimulator.step()`
情绪映射规则（connection 纯 PIR 驱动、safety 受巨响/移动/黑暗/极端温度扣分等）	`life_wild.py` `WildTargetGenerator.compute()`
计划四件套 vs 代码六传感器八通道	`AL-M26` 帖 vs `sensor_bridge.py` `CH_NAMES_WILD`
两版固件：Mega 2560 → UNO，输出协议一致	`sensor_firmware.ino` / `sensor_firmware_uno.ino`
全程 "mode": "sim"，从未跑过真实硬件	`wild_log.jsonl` 每行 mode 字段

Files and what they cannot tell you

These are the working files behind the wild-type experiment: firmware, the sensor bridge, the emotion core, the symbol tracker, and the logs from both the capped two-year run and the archived six-year one.One limitation applies to all of them, and I want it stated here where the files are rather than buried inside a reflection card. The log rotates. It holds roughly the last few thousand frames and overwrites everything before that. So for each run what survives is approximately the final ten minutes, and nothing in between.This is why AL-I16 can state that the coupling between physical presence and the connection dimension was absent at two virtual years and clearly alive at six, but cannot say when it appeared. That gap is in my instrument, not in the creature. Anyone reading these files should know which of the two they are looking at.

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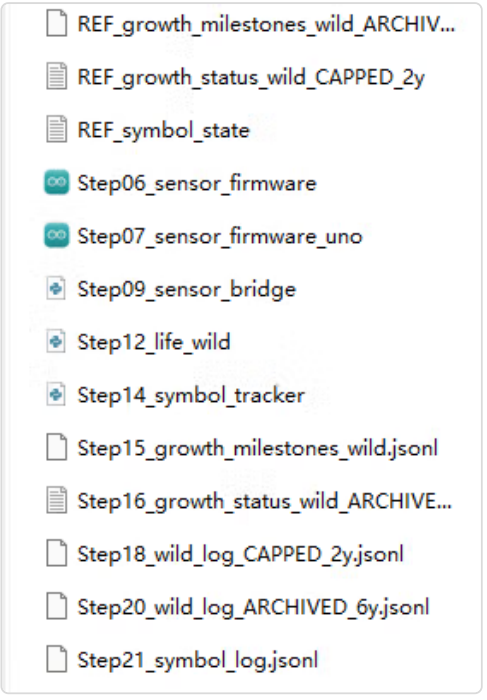

REALISE

The Problem I Kept Bumping Into In Plan A

I'd already built the two-track comparison by this point — Plan A with no language and no caregiver at all, just a blank randomly-initialised network wired to raw physical sensors; Plan B with an LLM caregiver raising the entity inside a simulated apartment, giving it a social world to grow up in. The whole reason for running both side by side was to isolate one question: what does social and linguistic experience actually add to a cognitive system that starts from literally nothing?

Sitting with Plan A specifically, there's a problem I kept bumping into without quite naming it. If the entity in Plan A has no language, how would I — or anyone — ever know it had developed any internal state at all? It can't tell me. The only evidence would have to be some form of expressive behaviour it invents on its own: a pattern in how it moves, signals, responds, that a person could learn to read, without that pattern ever being translated into words first.

That's the actual technical bet underneath Plan A: legible state without language. At this point I hadn't connected it to anything outside the experiment.

ID
RE-01

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Every Obvious Product Direction Was A Dead End Twice Over

Moving into Realise planning, I went looking at last year's IDE portfolios for direction, and kept getting stuck on the same frustration. Every obvious candidate for a physical output — a comfort object, a therapeutic toy, an emotional-support pet-like thing — was fundamentally about a human's feelings.

That's also exactly the territory my Unit 2 project sat in already (the ADHD therapeutic toy). So it felt like a dead end twice over, not just once: thematically repetitive, and interdisciplinarily thin.

I want to keep this in the record rather than skip to the answer, because the jump that followed didn't come from searching for a bridge between the experiment and a product. It came as a side effect of actively trying to avoid the companion framing.

ID
RE-02

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The Two Problems Are The Same Problem

Somewhere in the middle of rejecting those directions, it clicked.

The unsolved problem sitting right there in Plan A — how do you make an entity's state legible when there's no language mediating it — is the exact same problem as how does a person understand what a plant, a pond, or a room is experiencing, when none of them can tell you either.

I wasn't looking for this connection. I was trying to get away from something else, and this was what was underneath.

That reframed what the physical device would even be for: not something that expresses feelings to a person for the person's own sake, but something that makes an unreadable non-human system readable — with the entity itself as the translating medium, not the point.

ID
RE-03

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Why It Had To Be Universal, Not Fixed To One Biome

Once that clicked, I pushed it further than "translate one specific ecosystem."

Plan A and Plan B were never a single condition. The entire design was a comparison — two different rearing conditions run in parallel, specifically so that nothing about the result could be blamed on one fixed setup.

That same instinct — don't lock onto one fixed case, build something that generalises across conditions — is what made a device tied to one specific plant, or one specific fish tank, feel wrong as soon as I considered it. If the experimental design's whole point was testing across conditions, a product that only ever worked in one condition would have thrown away the most interesting part of what I'd already built.

So the "universal / chameleon" requirement isn't a separate aesthetic preference that appeared out of nowhere. It's the same design habit from the experiment, reapplied to the product.

ID
RE-04

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I Caught Myself Smuggling The Companion Framing Back In

There was a real detour here, worth keeping rather than tidying away.

My first instinct — even after landing on "translate an environment instead of a human" — was still to describe the interface in human emotional terms. I said I wanted it to work through human feeling.

I caught myself on this and deliberately widened it to biological emotion instead. Specifically because defaulting to human emotional categories would have quietly smuggled the companion framing back in through the side door, just dressed differently — the object would still ultimately be organised around a human's emotional experience, only with an ecological label on top.

That correction mattered as much as the original jump did.

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RE-05

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The Pivot: From "Training My Own AI Companion" to "Ecological Environment Translation"

Starting point: the entity's core sensing/emotion architecture already existed from Investigate (CfC + AutoNCP, five-dimensional emotional state). The open question for Realise was what this architecture should be for — what physical product it should become.

The instinct that was explicitly rejected: framing the output as an emotional-companionship device — "training your own AI companion," in the language originally used to describe it. This framing implicitly made the user's emotional experience the point of the device: something to project affection onto, in the electronic-pet tradition.

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RE-06

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Flipping The Order: Materials First, Function Second

From there the thinking got more concrete and less associative, in a way that felt like a shift in gear.

Instead of asking what the device should do and then hunting for hardware to match, I flipped the order: what can commodity sensors and a small local network actually survive and measure at all — temperature range, waterproofing, whether it needs a live language model or can run offline — and let that set the honest boundary of what the device could claim to do.

That habit isn't new either. It's the same posture Investigate was already built around when documenting what Plan A and Plan B could and couldn't demonstrate: state the scope honestly, don't claim more than the architecture can actually validate.

What this constraint killed, specifically: wool felt. AL-C5 proposed soft, hand-made organs driven by servos, and I was serious about it. But asking what commodity sensors can physically *survive* answers it immediately — this device's organs are probes going into soil and standing water, and a soft absorbent material cannot house that. Worth keeping the order of events straight: the material did not fail on its own terms. It was ruled out by the branch I took. Felt lost to a scope decision, not to a test.

ID

RE-07

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One Habit, Showing Up Three Times

Looked at end to end, the chain isn't really a series of separate decisions.

It's one intellectual habit — comparative rather than fixed to a single case; interested in legible expression without imposing a pre-built vocabulary on it; honest about what the architecture can actually prove — showing up three times:

once in how the twin-track experiment was designed, once in refusing
the companion framing, once in how the physical device's scope was set.

The physical object is less an "invention" than the same research disposition landing on a new surface.

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RE-08

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"Universal" Does Not Mean "Mediocre At Everything"

The obvious objection to a universal device is that it becomes a jack of all trades, master of none. That objection is usually correct — but it's worth being precise about why general-purpose devices normally end up mediocre, because this architecture avoids both of the usual failure modes.

General-purpose devices go mediocre at one of two layers:

Compromise at the sensing layer — using one "does-everything" sensor
to approximate several measurements, so nothing is measured accurately.
Averaging at the interpretation layer — decent hardware, but a single set of thresholds
stretched to cover soil, water and air at once, so it fits none of them precisely.

Both failures share one mechanism: taking an average. Averaging is what kills specialisation.

This design does neither.

The sensing layer is a union, not a compromise. Seven dedicated sensors are all physically present, each doing its own job. The pH probe here is the same physical component, at the same accuracy, as the pH probe on a dedicated water-quality meter — it does not become worse because a light sensor happens to be mounted beside it. The cost paid for this is size, expense and power draw — not accuracy. That distinction matters: there is a real cost, but the cost is not "measures things badly."

The interpretation layer switches, it does not average. When the tomato genome is loaded, what runs is a weighting and comfort-band set that is 100% specialised for tomato — nothing in it has been diluted to also accommodate water or air. The freshwater genome is not the mean of three scenarios; it is itself the freshwater specialist.

So the accurate description is not "a generalist." It is: a carrier capable of holding multiple specialists, activating one at a time. The analogy is an interchangeable-lens camera body — the body is not a compromise lens; it is an interface that lets you mount exactly the right lens each time.

ID
RE-09

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In One Specific Respect, It Is More Specialised Than A Dedicated Instrument

This is the part of the argument that goes on the offensive rather than merely defending.

A commercial soil monitor ships with generic thresholds for the category "soil," because the manufacturer cannot know where it will be sold or what soil it will be pushed into. It cannot be otherwise: a mass-produced instrument cannot know its deployment site in advance.

This device, through auto-calibration during its initial deployment period, learns what normal looks like for this particular pot, on this particular windowsill, in this particular season.

That means it is not only species-specialised but site-specialised — something a mass-produced dedicated instrument is physically unable to do. On the specific axis of "fit to this actual environment," a universal device here is more specialised than a dedicated one, not less.

There is a second capability that no dedicated instrument has: because channels with zero weight are still logged rather than switched off, the device retains data from outside its current genome's frame of reference. A soil monitor can never tell you "conditions here have gone beyond what soil semantics can describe," because it has no other channels to consult. This device retains all peripheral data, and can in principle detect that the wrong genome is loaded for the actual environment. That is a capability beyond the specialist, not a shortfall against it.

The biological analogy holds structurally, not just rhetorically: vertebrate nervous systems are highly conserved across species — the real difference lies in which sensory organs are attached and how the channel weightings are tuned. Neurons in an eagle's eye are not more advanced than a mole's; the substrate is general, the configuration is specialised. "A general brain with pluggable sensory perspectives" is not borrowing biology as a metaphor — it is the same structural principle.

ID

RE-10

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The Prototype Is Not The Ceiling: Separating Resource Limits From Architectural Limits

A statement worth making explicitly, because it determines what this prototype should and should not be judged on.

What this prototype exists to demonstrate is the architecture, not the product. Specifically: that swapping a genome configuration genuinely changes how identical sensor readings are interpreted; that the two-layer sensory weighting transform runs; that the emotional core maintains continuous state across time. Those are architectural claims, and a prototype can validate them.

What constrains the current build is resources, not architecture. This is a Graduate Diploma project: no team, no budget line, a five-week Realise window, and only components obtainable and deliverable within that window. That produces real limitations — fewer sensing channels than the concept ideally wants, an enclosure with 3D-printed rather than manufactured tolerances, runtime measured in hours rather than indefinite solar autonomy, a calibration period measured in days rather than seasons.

The distinction that matters: every one of those limitations scales with resources, and none of them requires the architecture to change. More channels means adding probes to the same array and extending the same genome schema. Longer runtime means the duty-cycled coprocessor power architecture already identified as the correct long-term answer. Longer calibration means leaving the device deployed for longer — the same code, more time.

So the vision should not be assessed against the fidelity of this build. A prototype constrained by a student's material access and timeline is evidence about what one person could physically assemble in five weeks. It is not evidence about the ceiling of the concept — those are two different questions, and only the first one is being answered here.

What would legitimately challenge the vision is an architectural objection: that genome switching does not meaningfully change interpretation, that the two-layer transform is the wrong abstraction, that the emotional framing does not aid comprehension over a numeric readout. Those critiques are welcome and are the ones worth having. Limitations traceable to budget, team size, or the length of the term are a different category, and are documented here so they are not mistaken for the first kind.

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RE-11

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Portfoilo refelcetion

Portfolio review session - favourite page

Image 1: Original layout.
Image 2: Everyone's suggestions.
Image 3: After revision.

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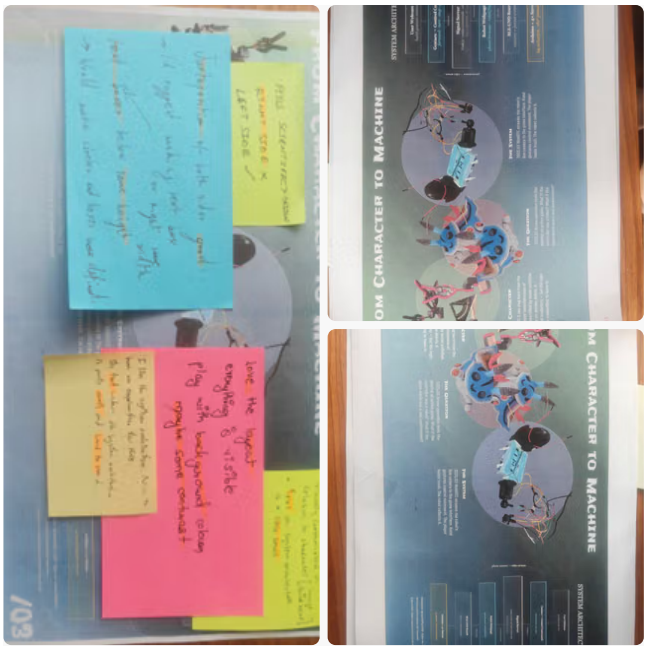

Portfolio review session - hardest page

Image 1: Original layout.
Image 2: Everyone's suggestions.
Image 3: After revision.

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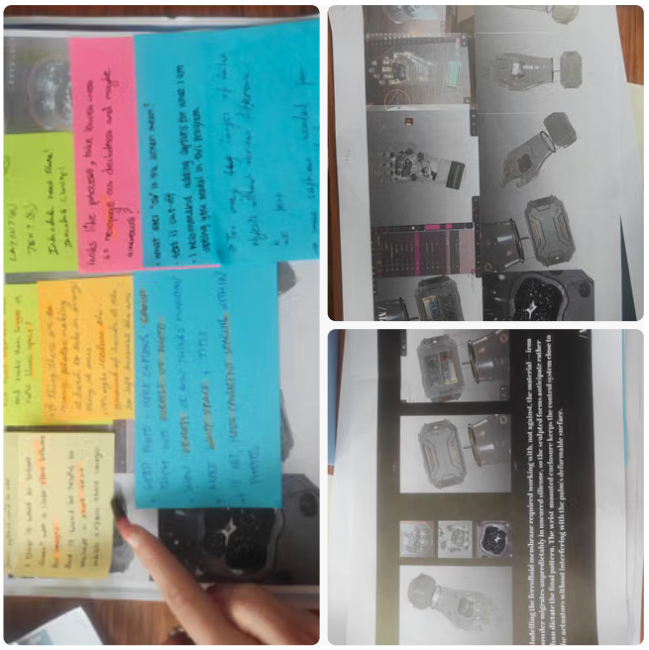

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personal reflection

I noticed that Western portfolios place a strong emphasis on concise information and white space, rather than the "noisy collage style" commonly used by Chinese students — dense with information and illustrations. Instead, they prefer to strip away unnecessary decorative elements. To me these are simply different aesthetic styles, but the minimal approach is better suited to a GD portfolio like this, where the volume of information is smaller and the focus is on hands-on making. So for me, the hardest part of going minimal might be **deciding which information to show out** of everything I have because I really do have a lot.

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PR-04

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